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Case Studies

Why look at case studies?

Outline

Classic networks:

- LeNet-5 ←
- AlexNet ←
- VGG ←

ResNet (152)

Inception

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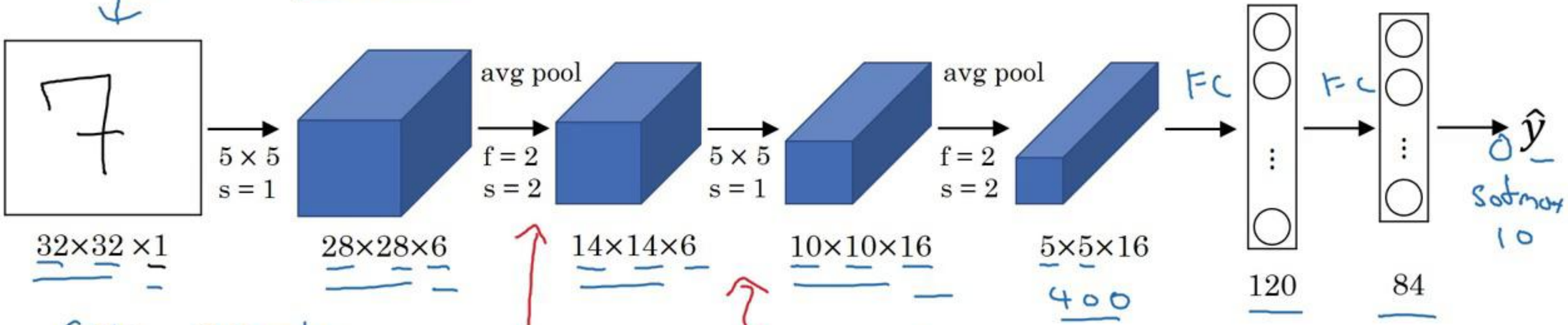
Case Studies

Classic networks

LeNet - 5



60K parameters.

non-linearity after pooling

$n_H \times n_W \times n_C$ $f \times f \times n_C$

$n_H, n_W \downarrow$ $n_C \uparrow$

conv pool conv pool fc fc output

Advanced: Sigmoid/tanh ReLU

II III

[LeCun et al., 1998. Gradient-based learning applied to document recognition]

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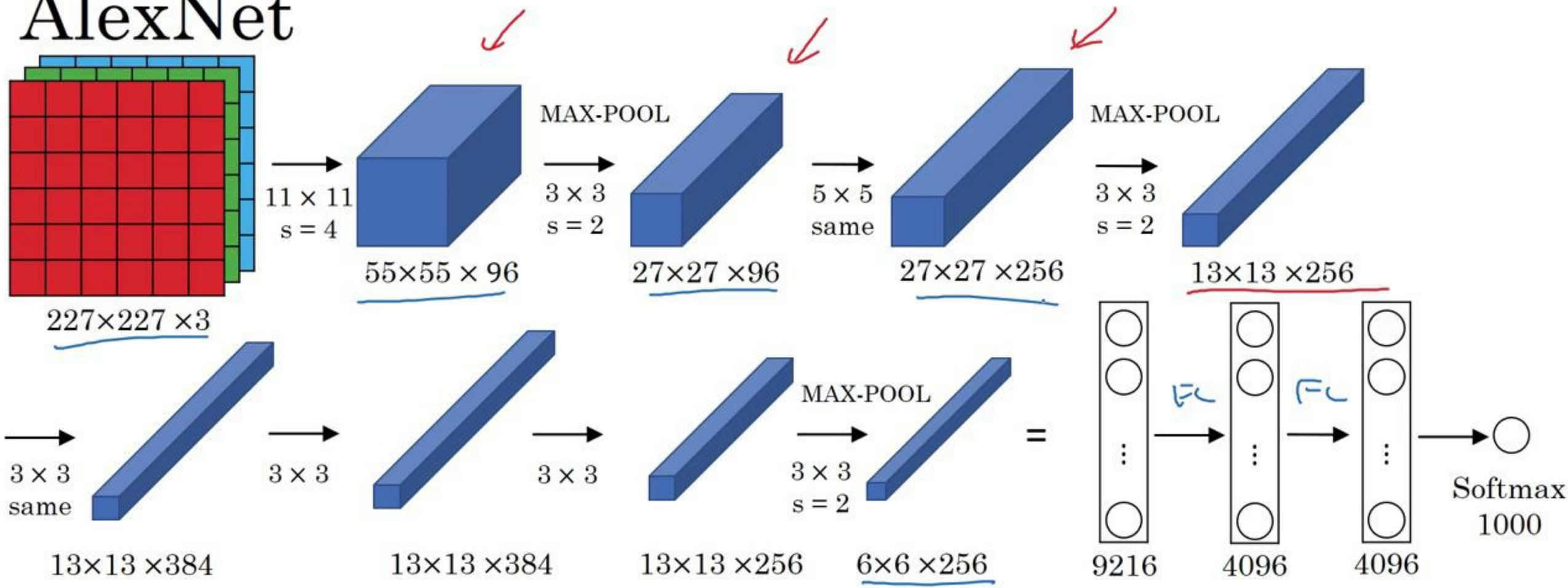
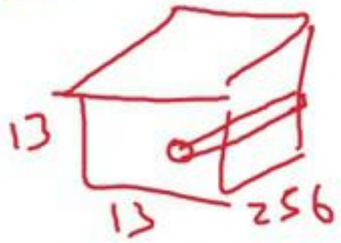
AlexNet



- Similar to LeNet, but much bigger. 9216 ~60M parameters
- ReLU
- Multiple GPUs.
- Local Response Normalization (LRN)



[Krizhevsky et al., 2012. ImageNet classification with deep convolutional neural networks]

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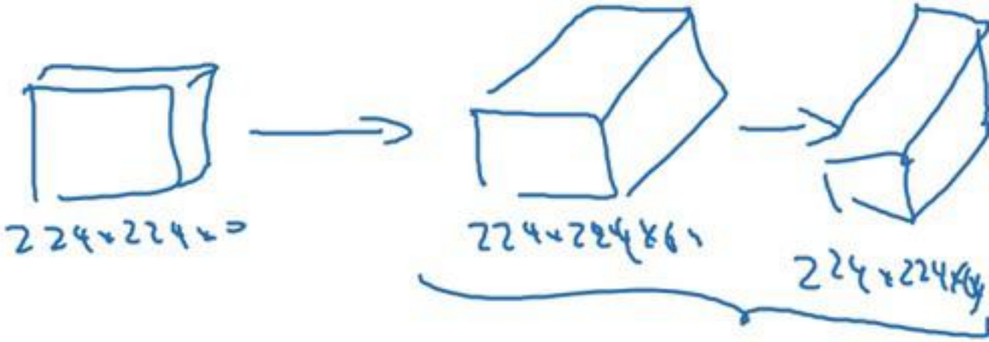
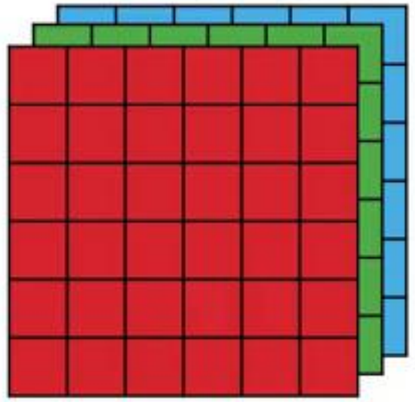
VGG - 16

VGG-19

CONV = 3×3 filter, s = 1, same MAX-POOL = 2×2 , s = 2





224×224 ×3

→ [CONV 64] ×2 → 224×224×64 → POOL → 112×112 ×64 → [CONV 128] ×2 → 112×112 ×128 → POOL → 56×56 ×128

→ [CONV 256] ×3 → 56×56 ×256 → POOL → 28×28 ×256 → [CONV 512] × 3 → 28×28 ×512 → POOL → 14×14×512

→ [CONV 512] ×3 → 14×14 ×512 → POOL → 7×7×512 → FC 4096 → FC 4096 → Softmax 1000

$n_H, n_W \downarrow$ $n_C \uparrow$ ~138M

[Simonyan & Zisserman 2015. Very deep convolutional networks for large-scale image recognition]

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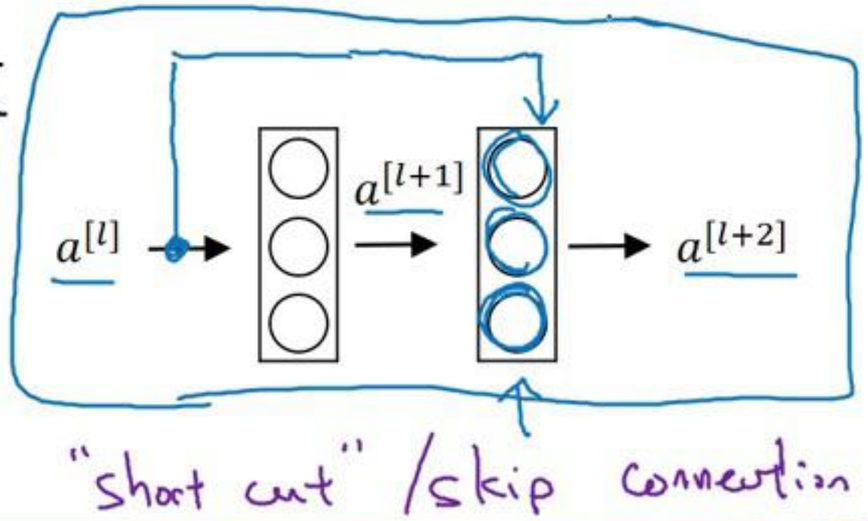
Case Studies

Residual Networks (ResNets)

Residual block



"short cut" /skip connection

$a^{[l]} \rightarrow$ Linear $\rightarrow$ ReLU $\xrightarrow{a^{[l+1]}}$ Linear $\rightarrow \oplus \rightarrow$ ReLU $\rightarrow a^{[l+2]}$

"main path"

$$z^{[l+1]} = W^{[l+1]} a^{[l]} + b^{[l+1]} \qquad a^{[l+1]} = g(z^{[l+1]}) \qquad z^{[l+2]} = W^{[l+2]} a^{[l+1]} + b^{[l+2]} \qquad \cancel{a^{[l+2]} = g(z^{[l+2]})}$$

$$a^{[l+2]} = g(z^{[l+2]} + a^{[l]})$$

[He et al., 2015. Deep residual networks for image recognition]

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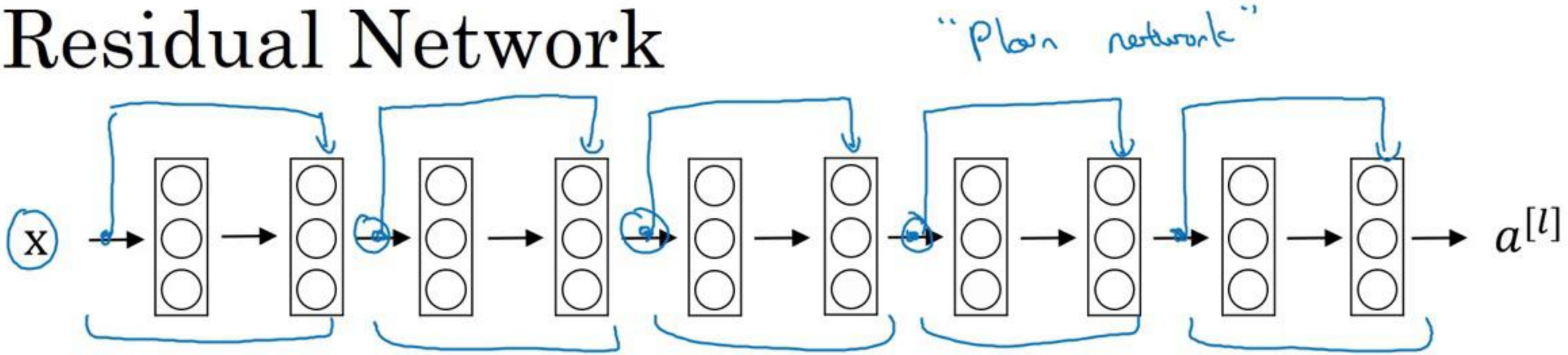
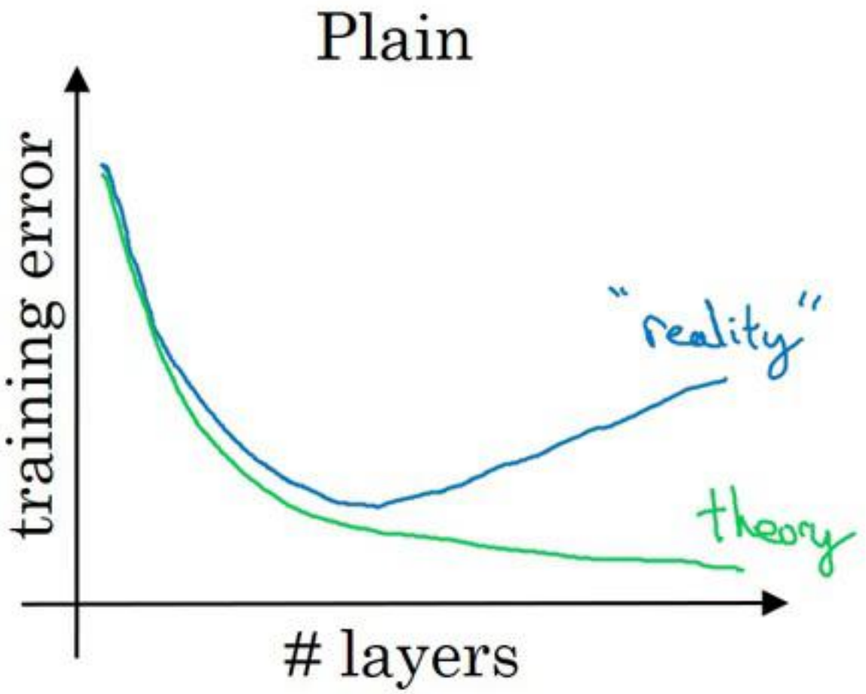
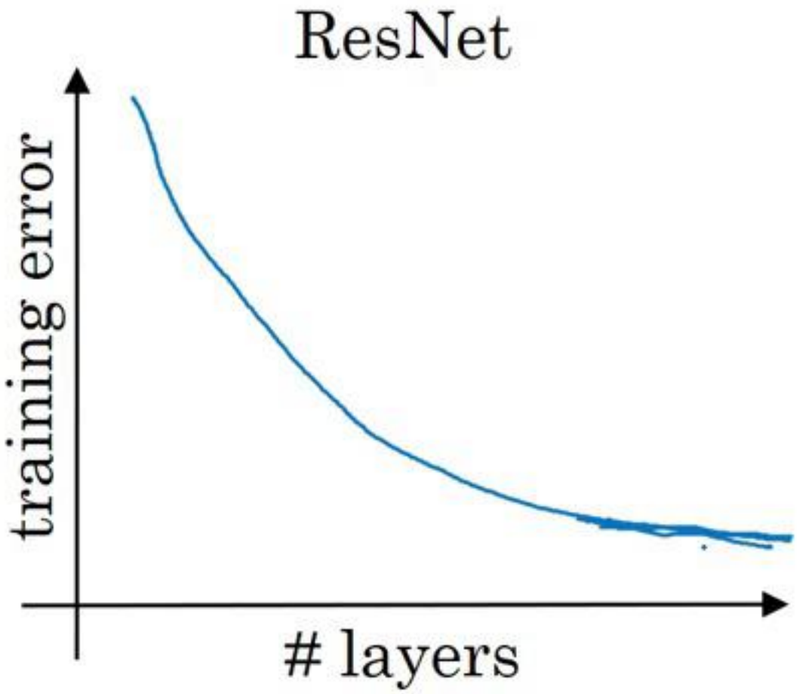
Residual Network







[He et al., 2015. Deep residual networks for image recognition]

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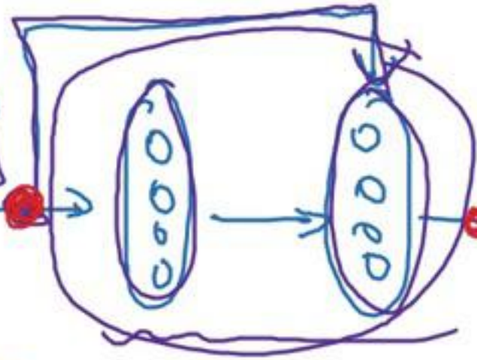
Why ResNets work

Why do residual networks work?

$X \rightarrow$ Big NN $\rightarrow a^{[l]}$

Identity function is easy for Residual block to learn!

$X \rightarrow$ Big NN $\rightarrow a^{[l]} \rightarrow$  $\rightarrow a^{[l+2]}$

"Same"

ReLU. $a \geq 0$

$$a^{[l+2]} = g(z^{[l+2]} + a^{[l]})$$

$$= g(\cancel{W^{[l+2]} a^{[l+1]} + b^{[l+2]}} + W_s a^{[l]}) = g(a^{[l]})$$

$$= a^{[l]}$$

256 $\quad$ $\mathbb{R}^{256 \times 128}$ $\quad$ 128

If $W^{[l+2]} = 0$, $b^{[l+2]} = 0$

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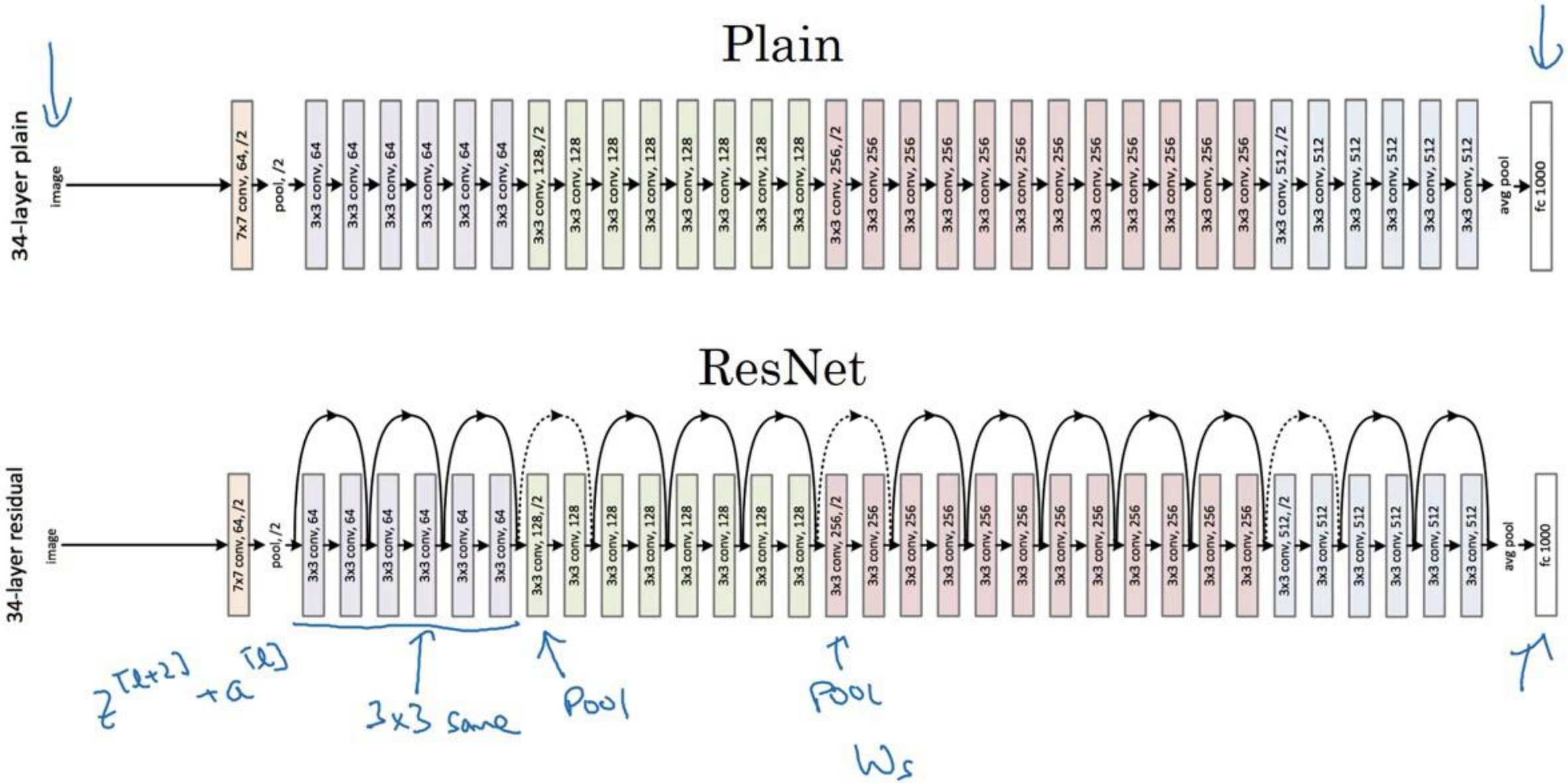
ResNet



[He et al., 2015. Deep residual networks for image recognition]

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Case Studies

Network in Network and 1×1 convolutions

Why does a 1×1 convolution do?

1	2	3	6	5	8
3	5	5	1	3	4
2	1	3	4	9	3
4	7	8	5	7	9
1	5	3	7	4	8
5	4	9	8	3	5

$6 \times 6 \times 1$

$*$ $\boxed{2}$ $=$

2	4	6	...		

[Lin et al., 2013. Network in network]

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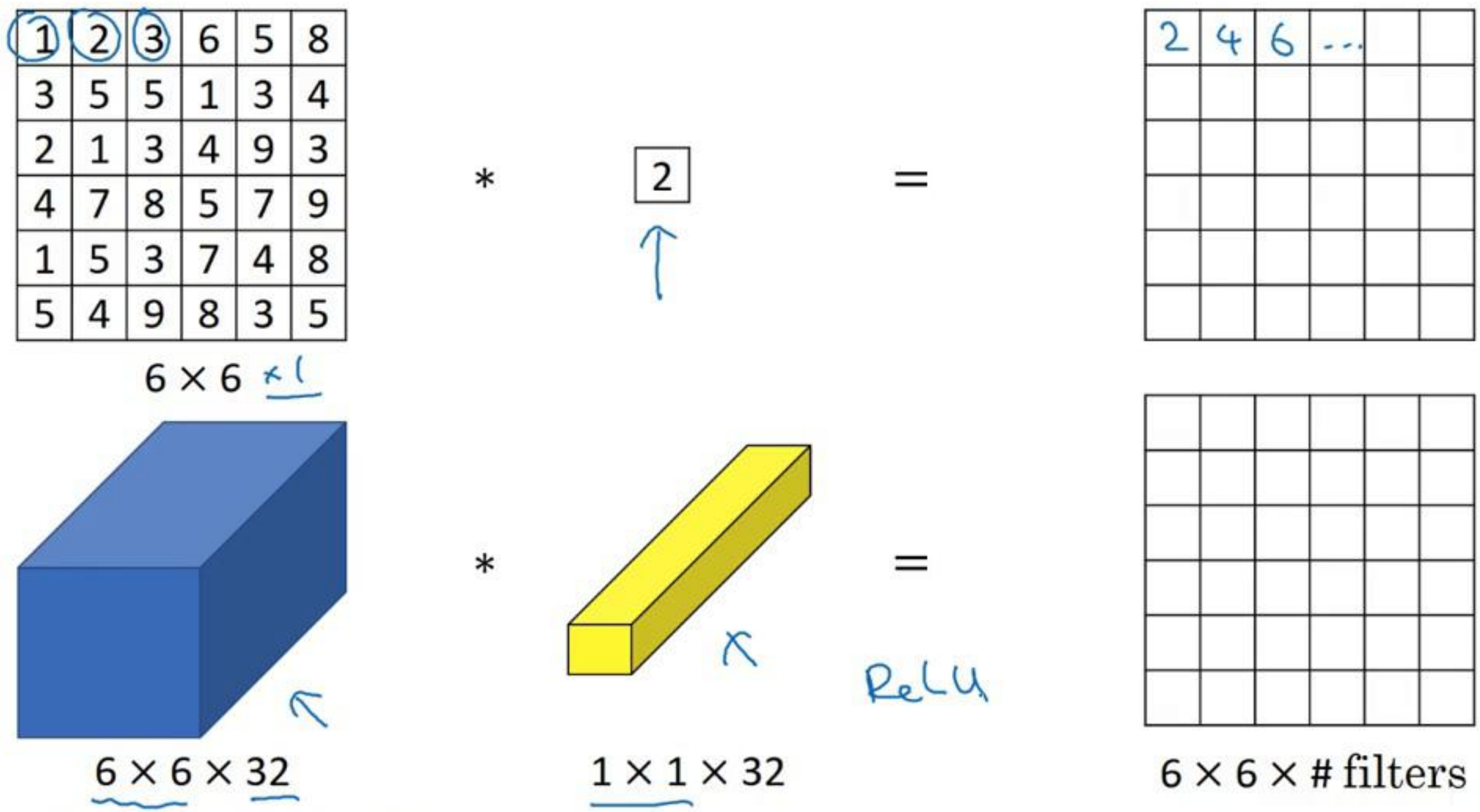
Why does a 1×1 convolution do?



[Lin et al., 2013. Network in network]

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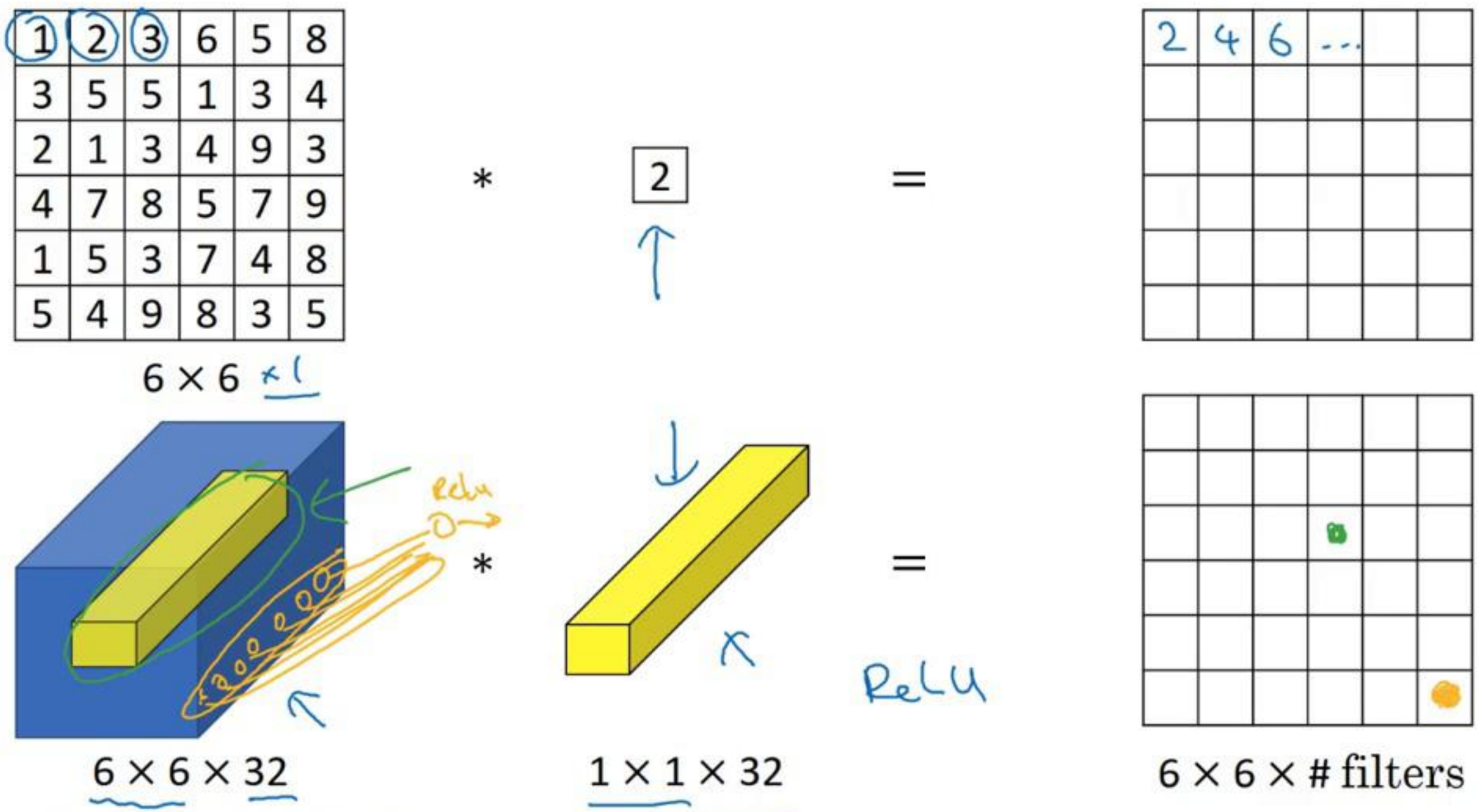
Why does a 1×1 convolution do?



[Lin et al., 2013. Network in network]

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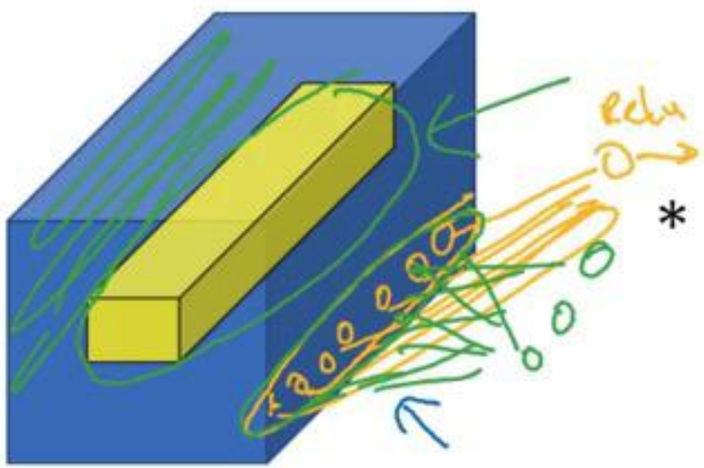
$6 \times 6 \times 1$

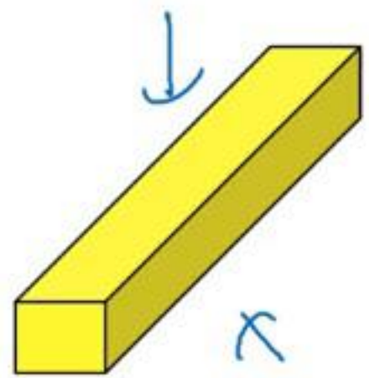
$*$ $\boxed{2}$ $=$

2	4	6	...		



$6 \times 6 \times 32$

$*$



$1 \times 1 \times 32$

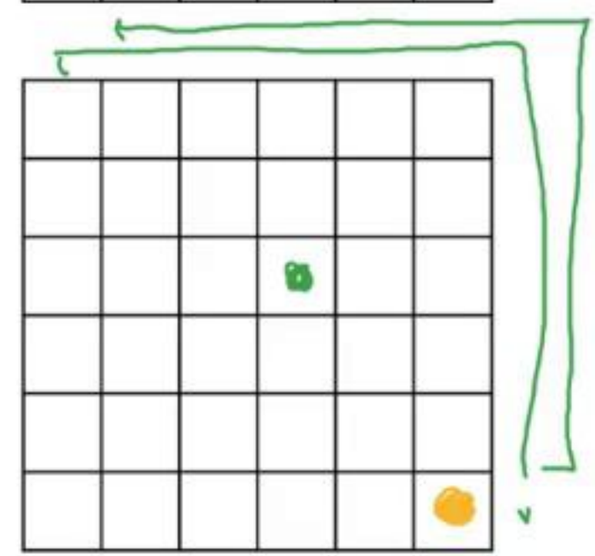
32 → # filters.

$n_c^{[l+1]}$

$=$

ReLU

Network in Network



$6 \times 6 \times$ # filters

[Lin et al., 2013. Network in network]

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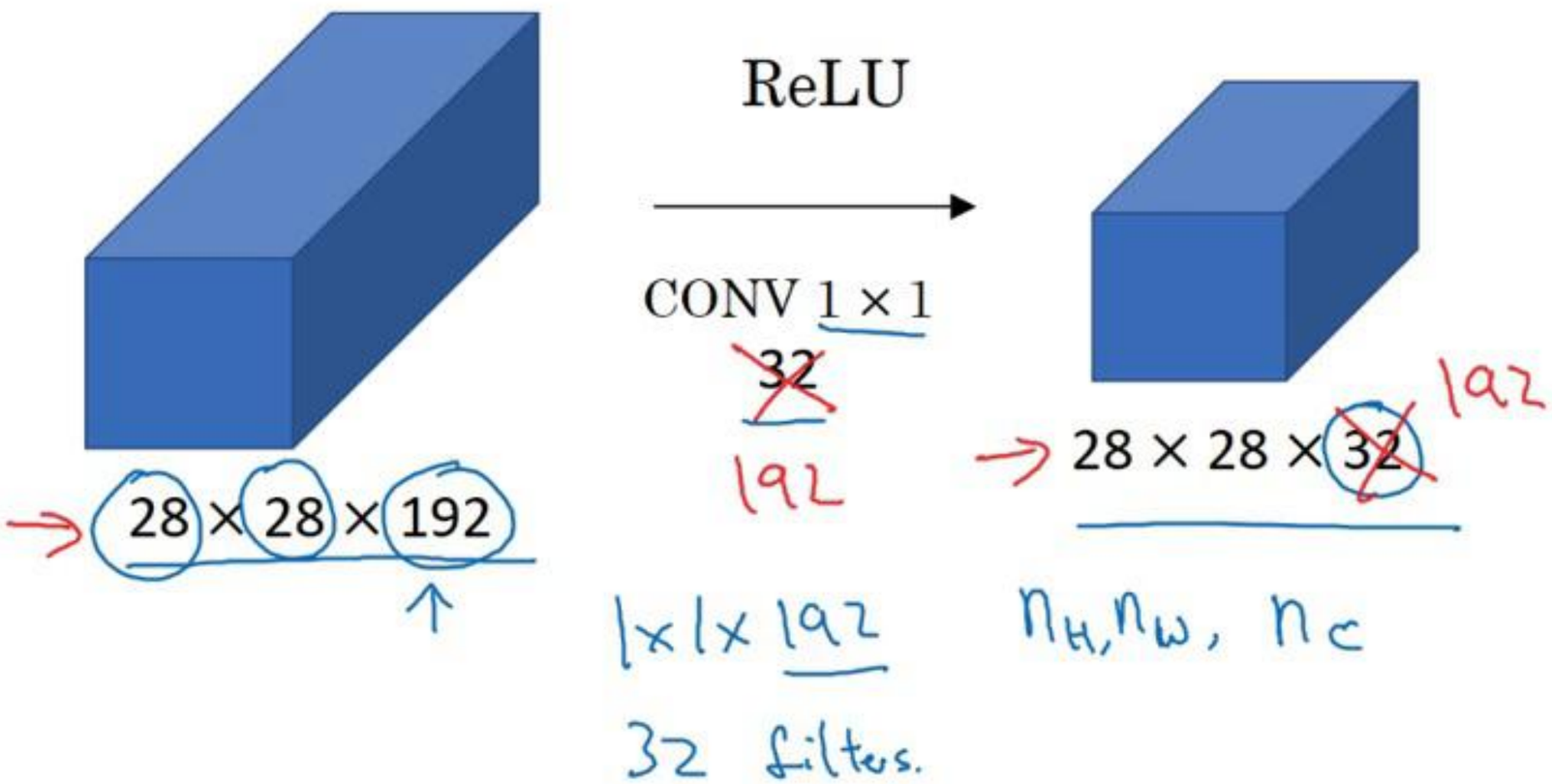
Using 1×1 convolutions



[Lin et al., 2013. Network in network]

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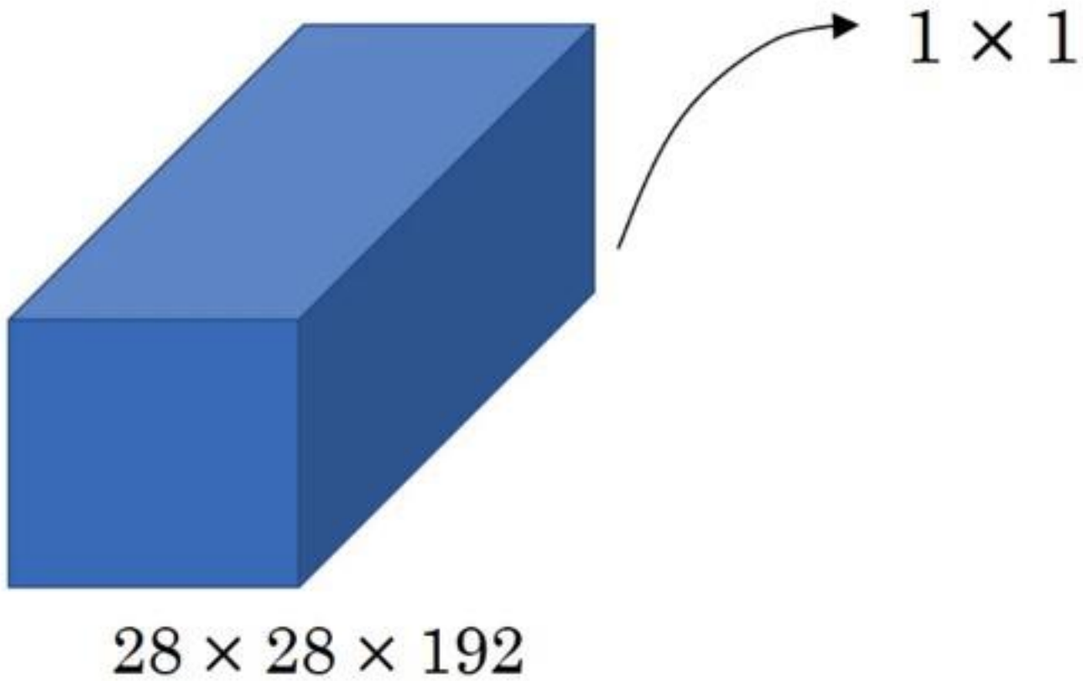
Case Studies

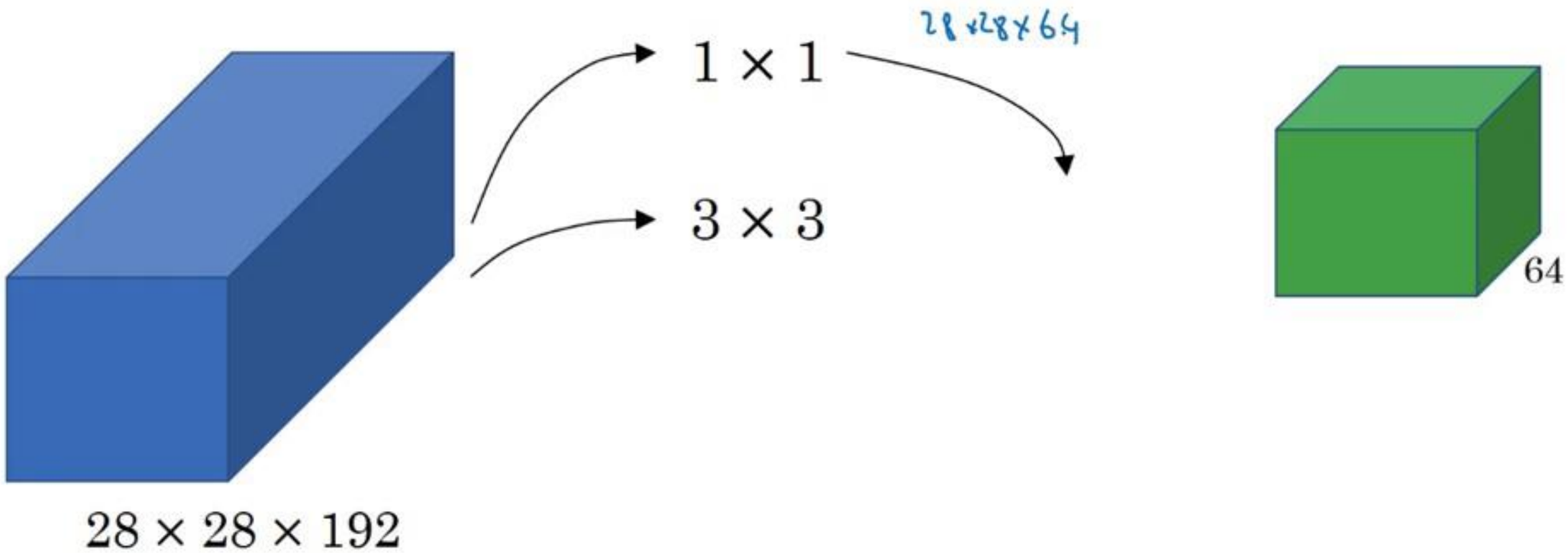
Inception network motivation

Motivation for inception network



[Szegedy et al. 2014. Going deeper with convolutions]

Motivation for inception network



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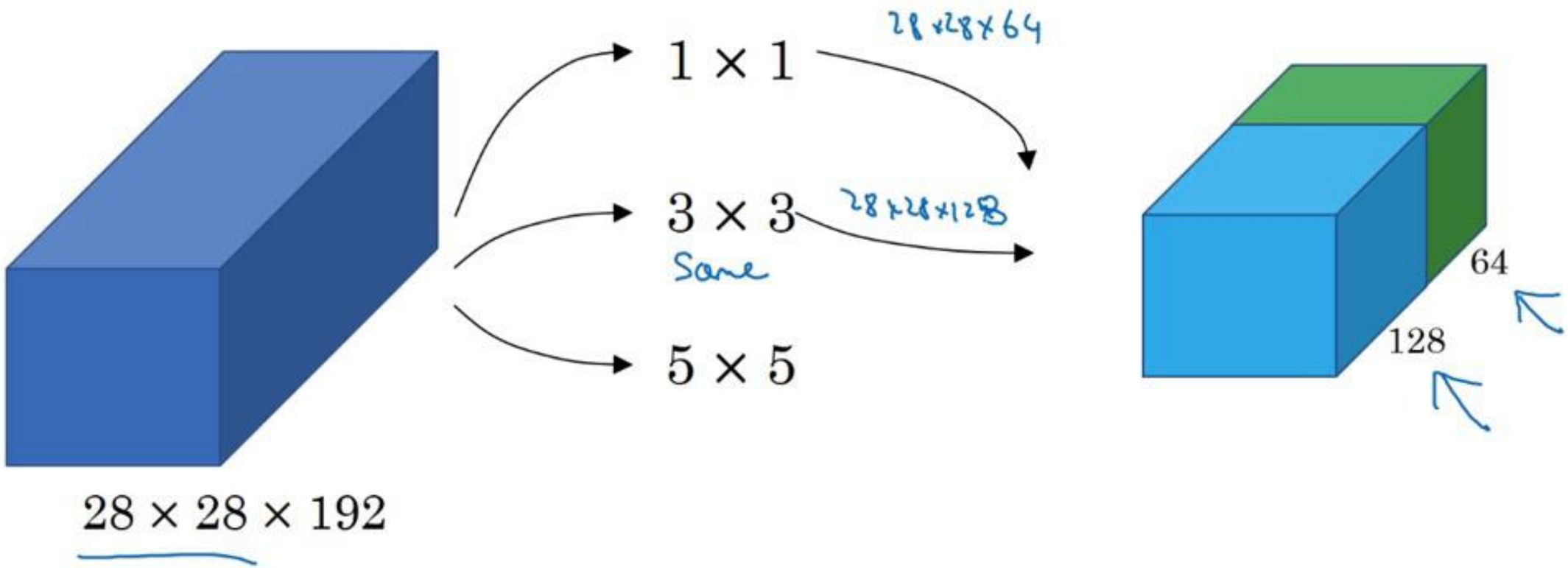

[Szegedy et al. 2014. Going deeper with convolutions]

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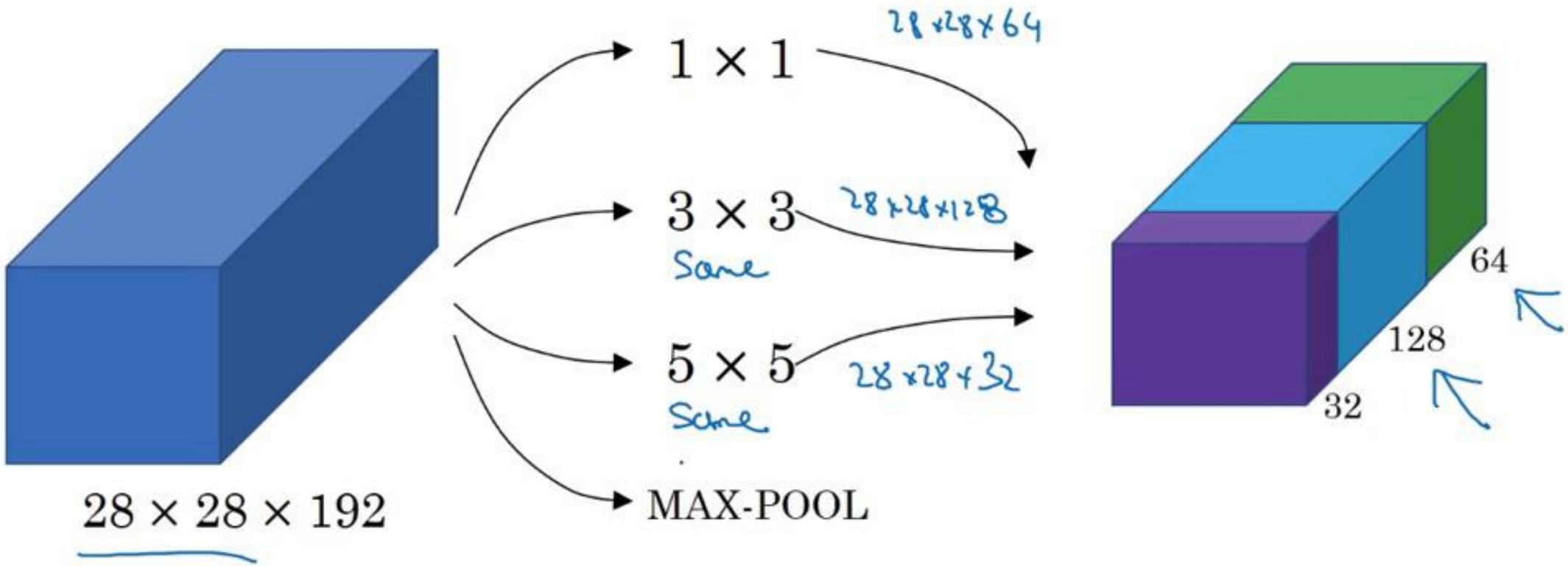
Motivation for inception network



[Szegedy et al. 2014. Going deeper with convolutions]

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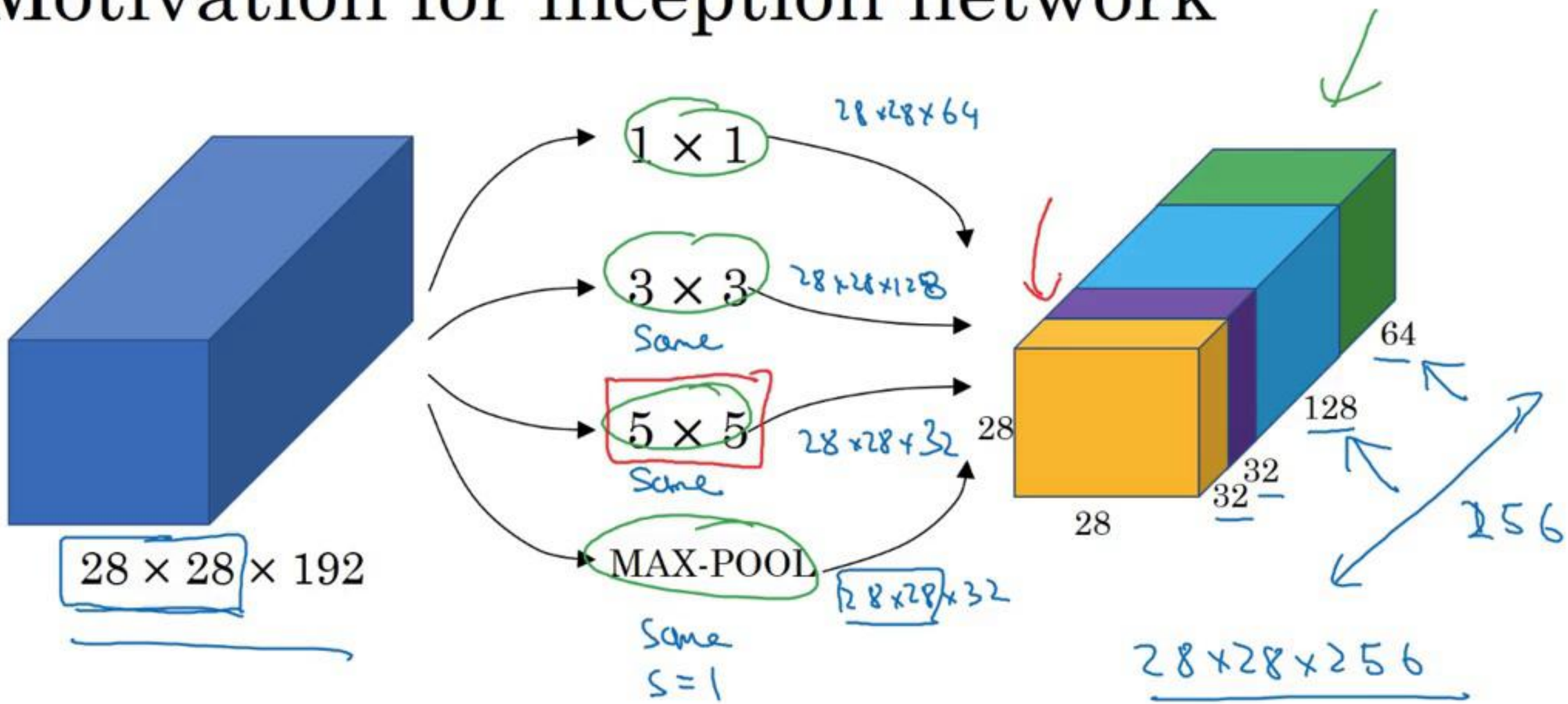
Motivation for inception network



[Szegedy et al. 2014. Going deeper with convolutions]

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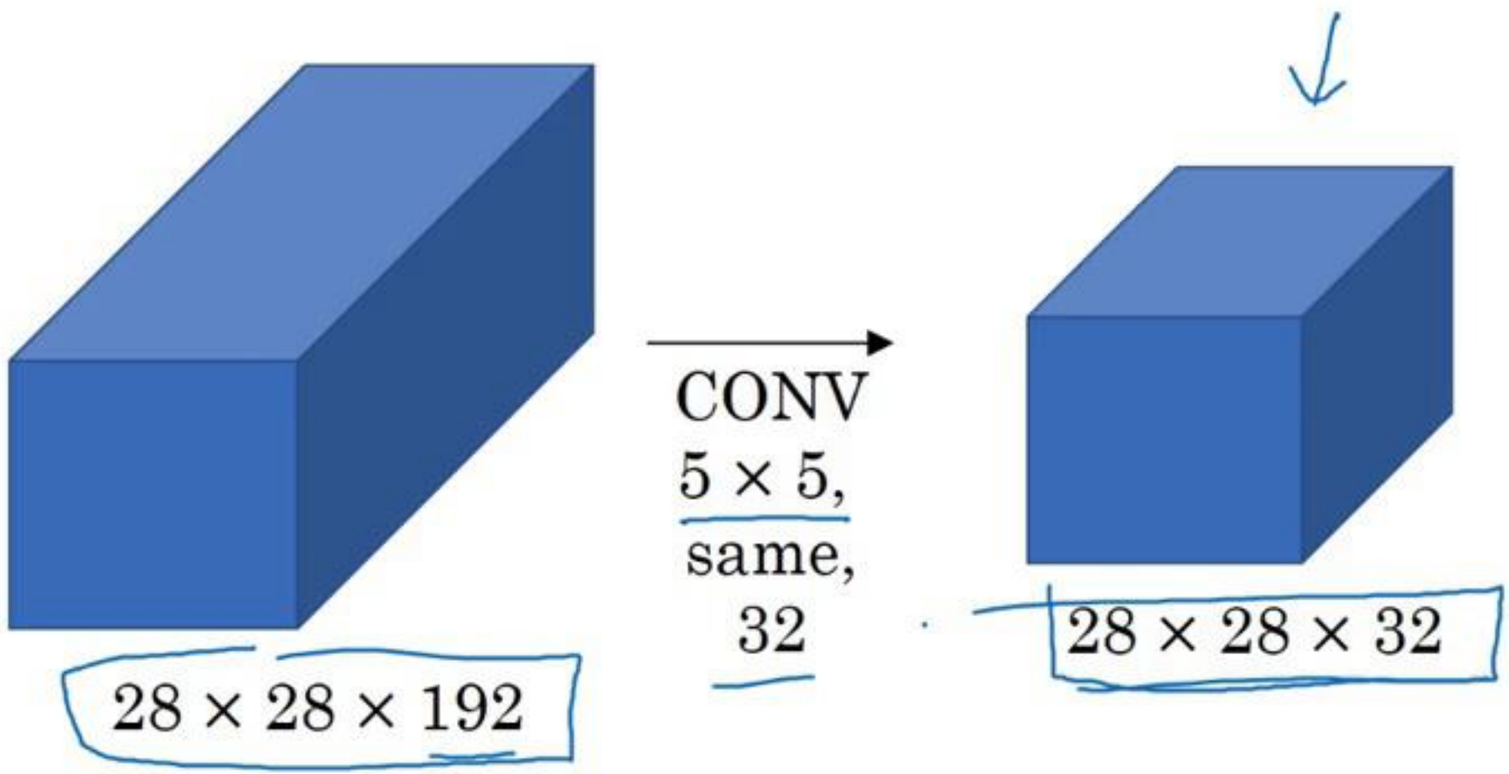
The problem of computational cost

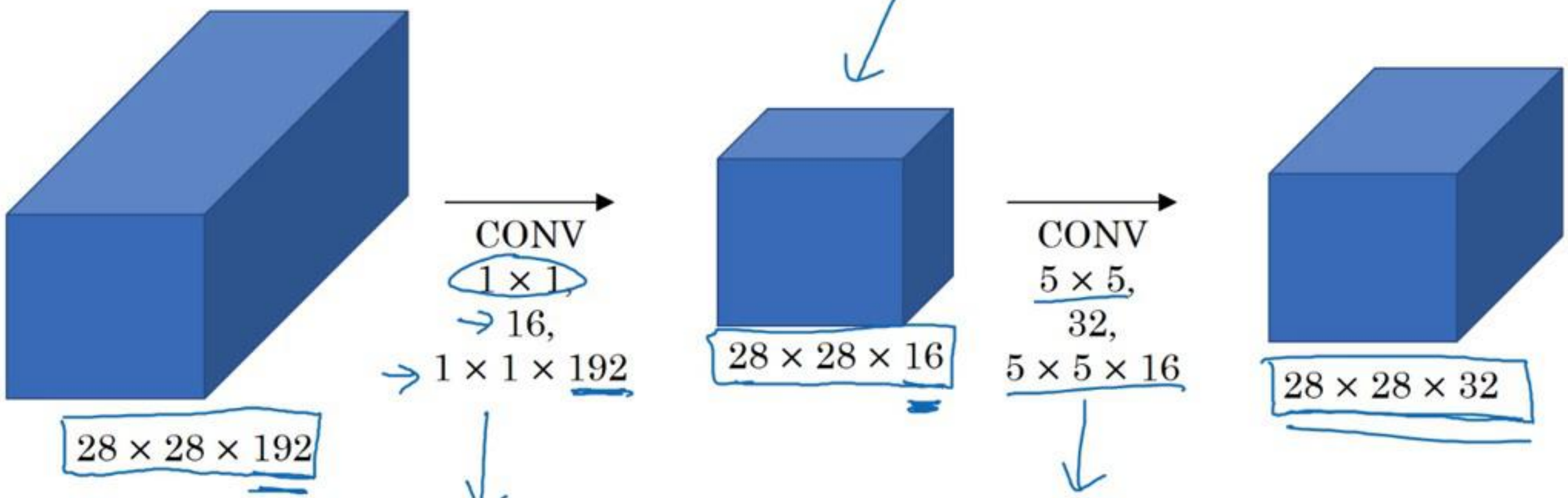


32 filters. filters are 5x5x192.

28x28x32 x 5x5x192 = 120M.

Using 1×1 convolution

"bottleneck layer"



$28 \times 28 \times 16 \times 192 = 2.4M$

$28 \times 28 \times 32 \times 5 \times 5 \times 16 = 10.0M$

$12.4M$

$120M$

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Case Studies

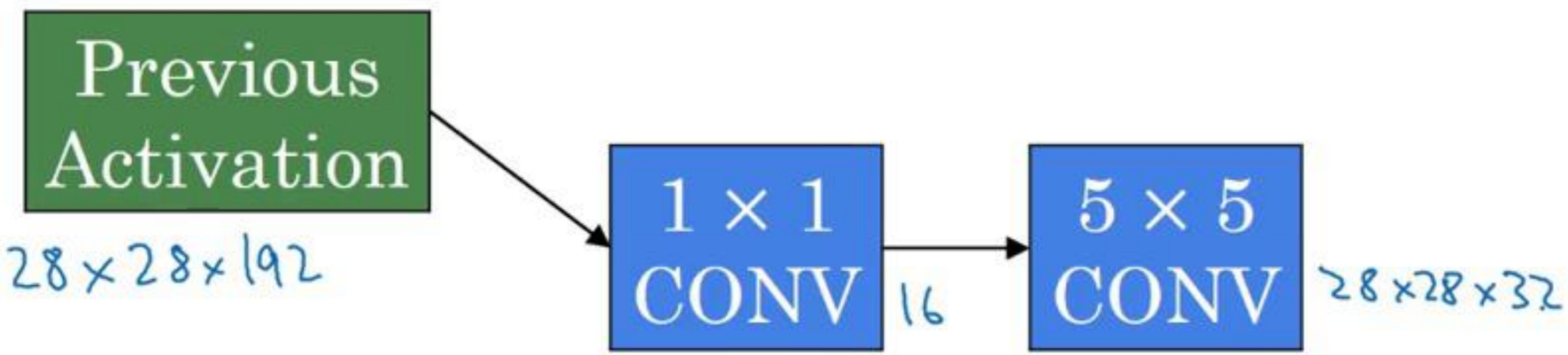
Inception network

Inception module

Previous Activation

$28 \times 28 \times 192$

Inception module



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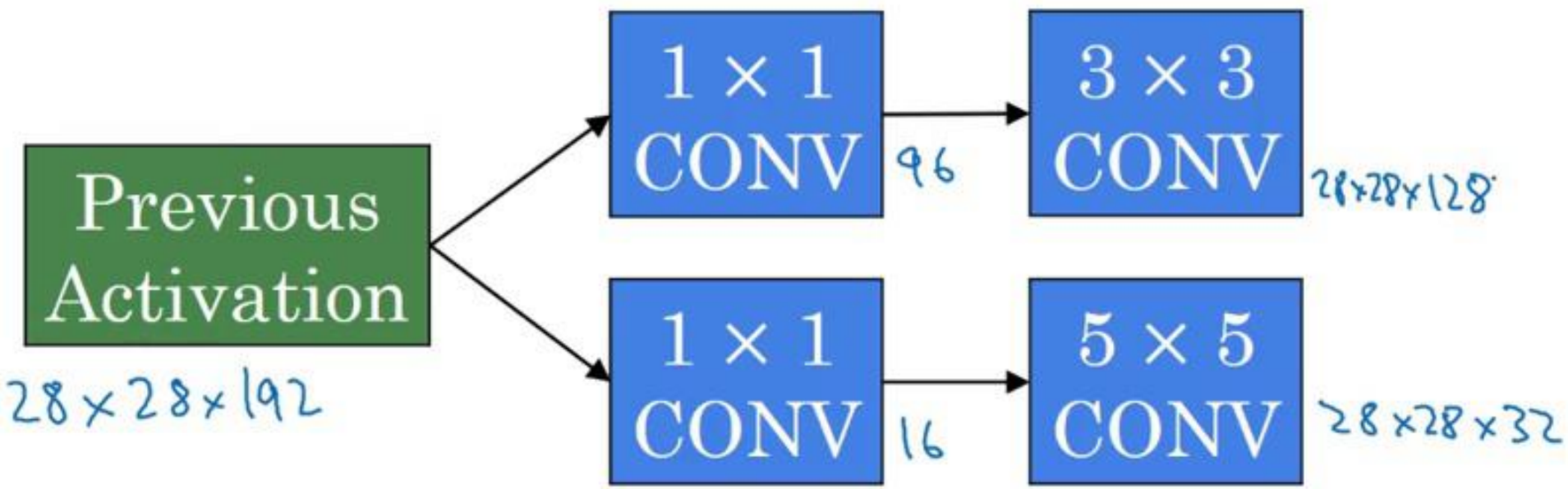
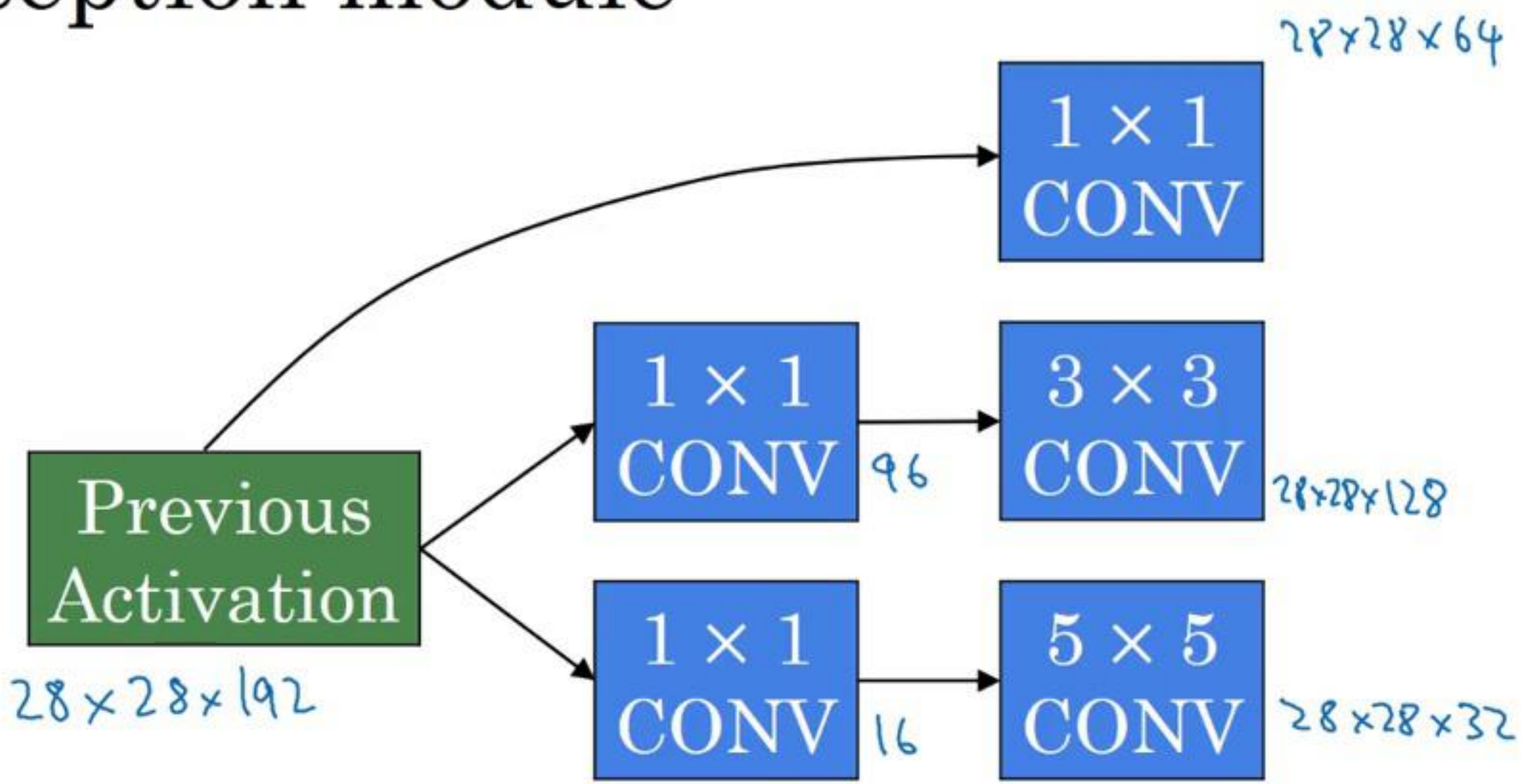
Inception module



Inception module



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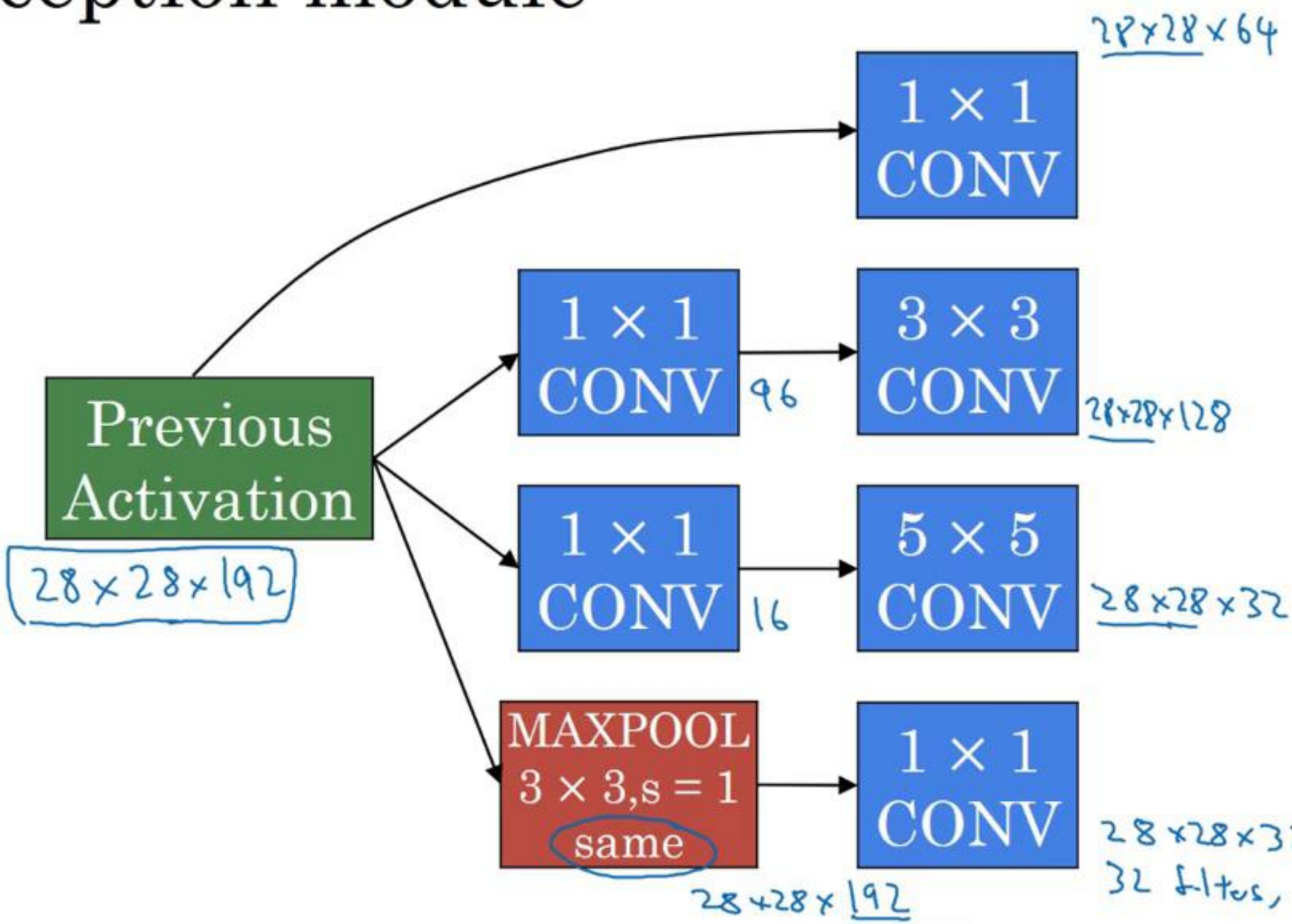
Inception module



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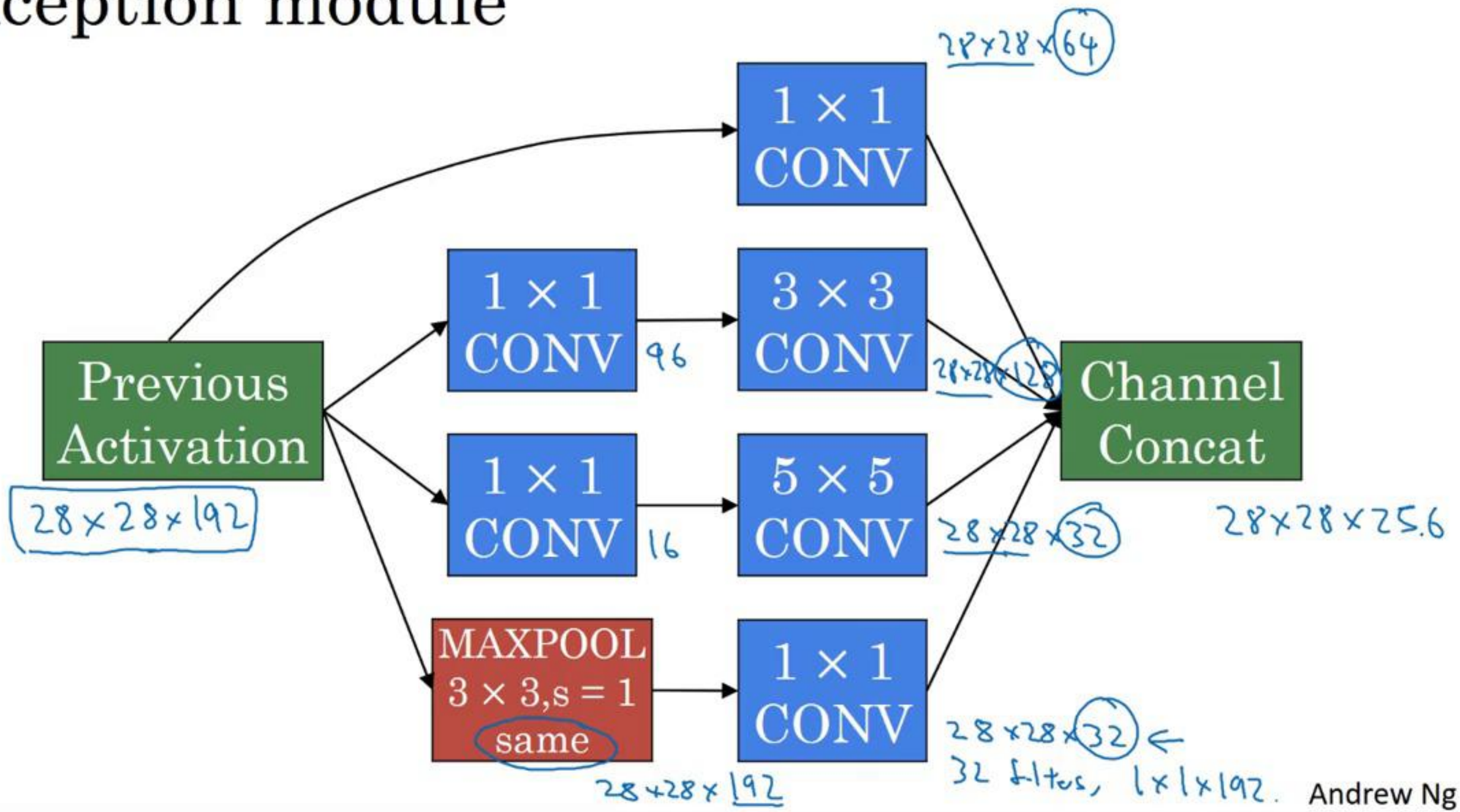
Inception module



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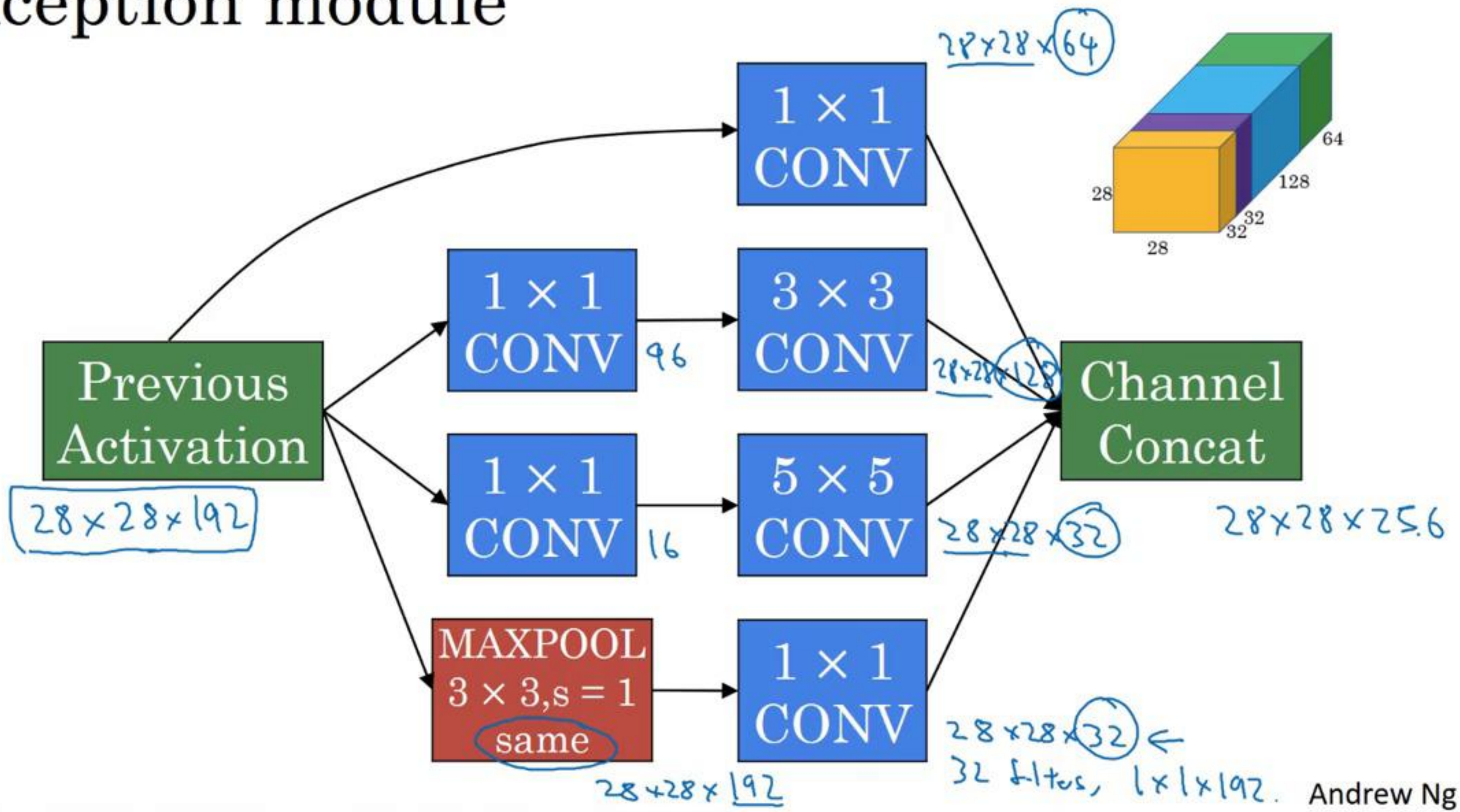
Inception module



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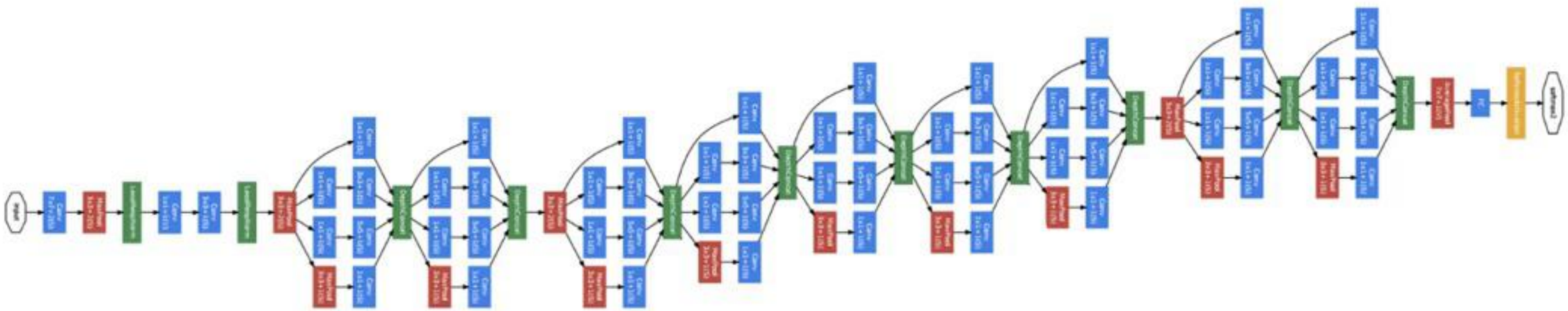
Inception network



[Szegedy et al., 2014, Going Deeper with Convolutions]

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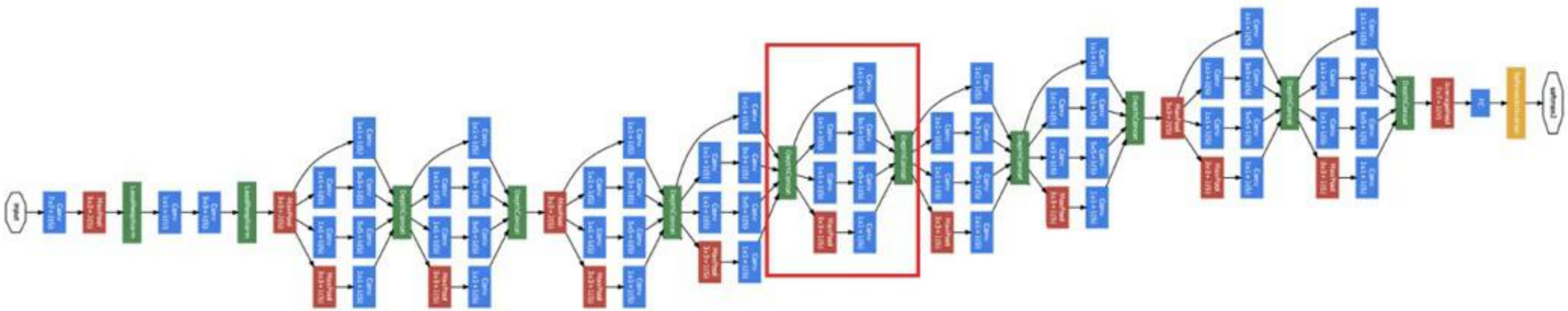
Inception network



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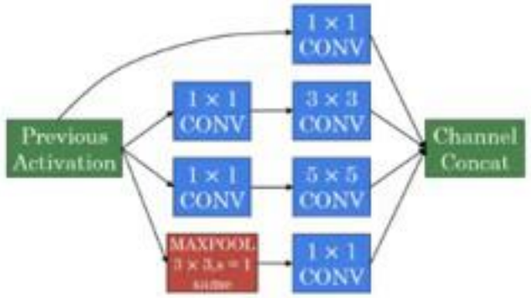
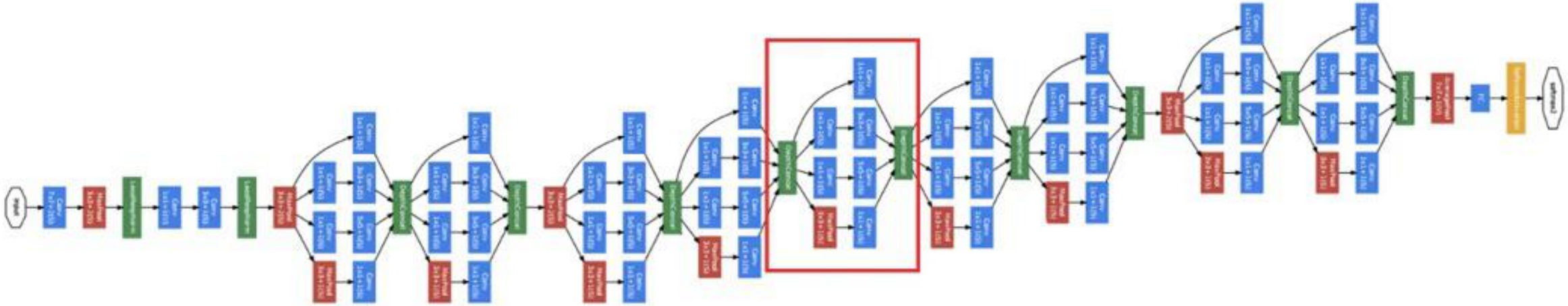
Inception network





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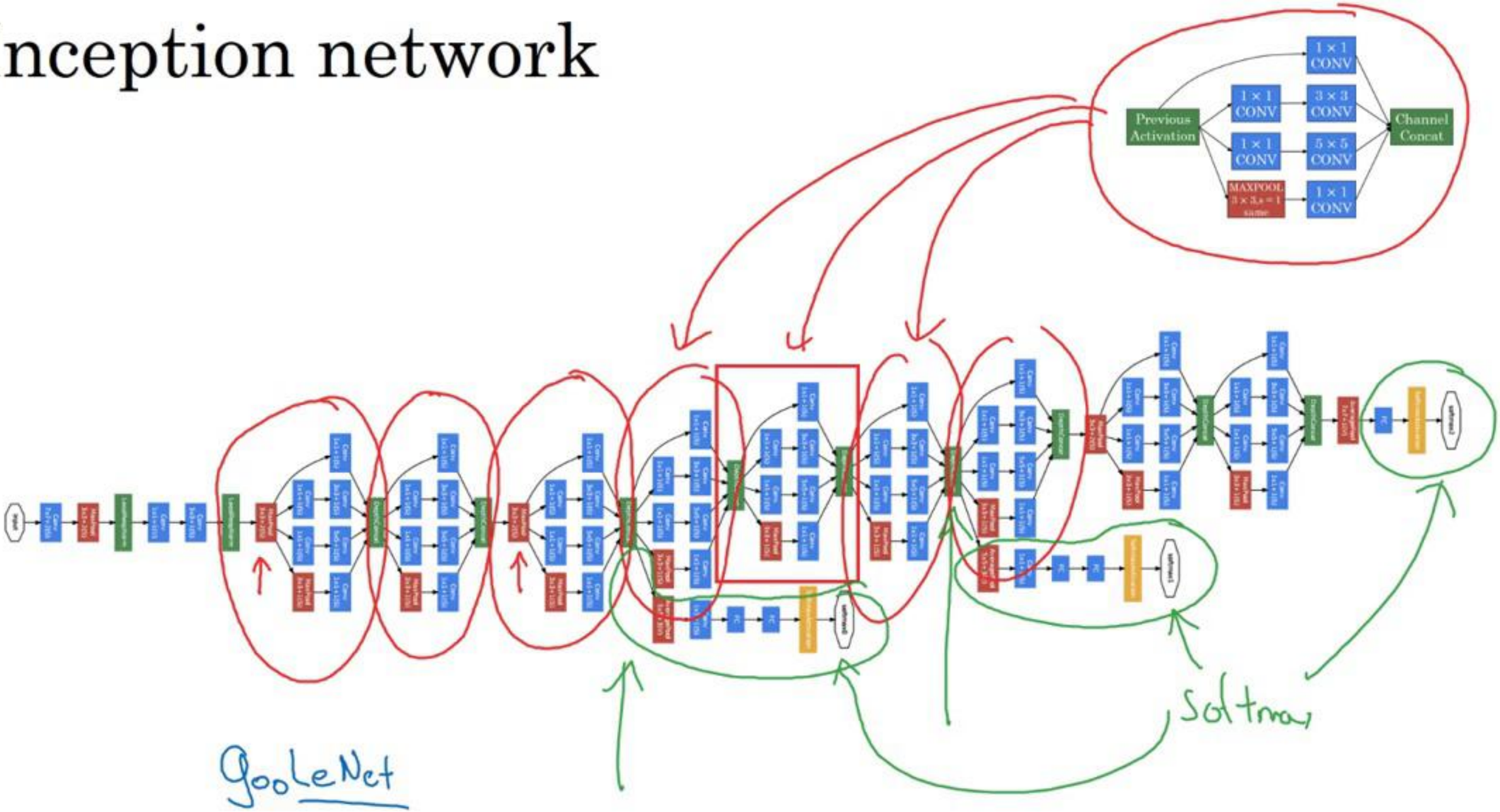
Inception network



[Szegedy et al., 2014, Going Deeper with Convolutions]

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Practical advice for using ConvNets

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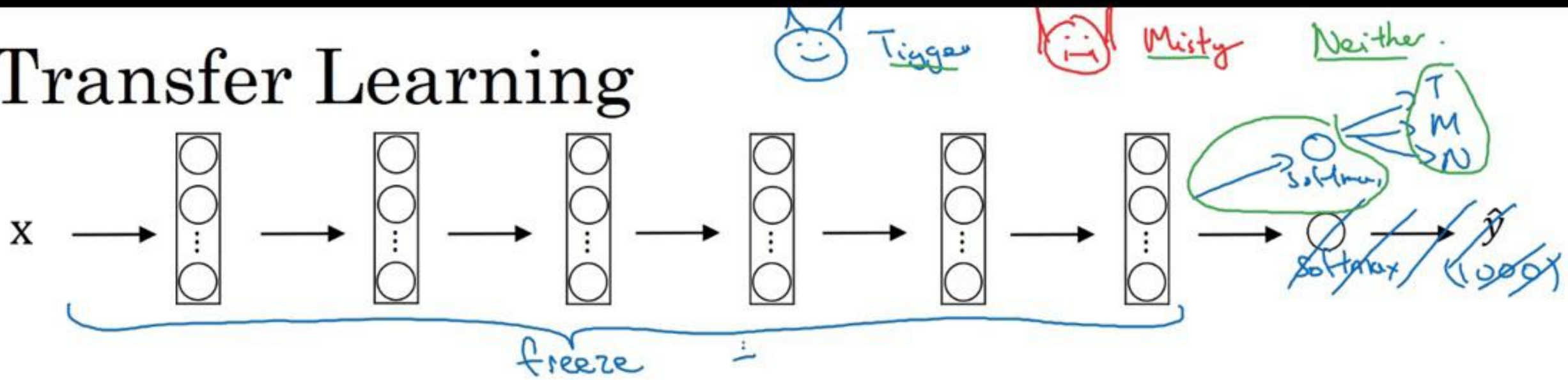
Transfer Learning

Transfer Learning



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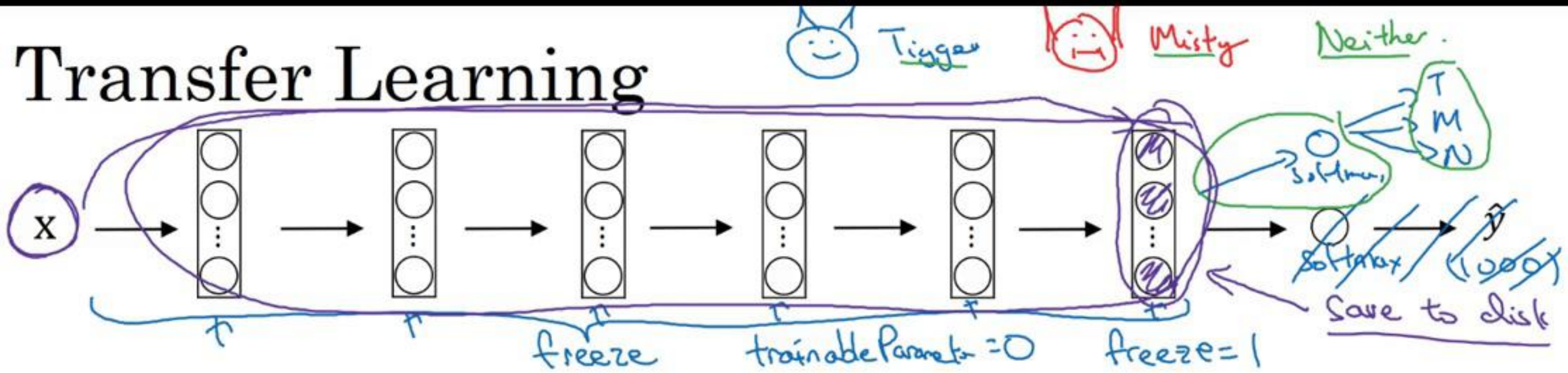
Transfer Learning



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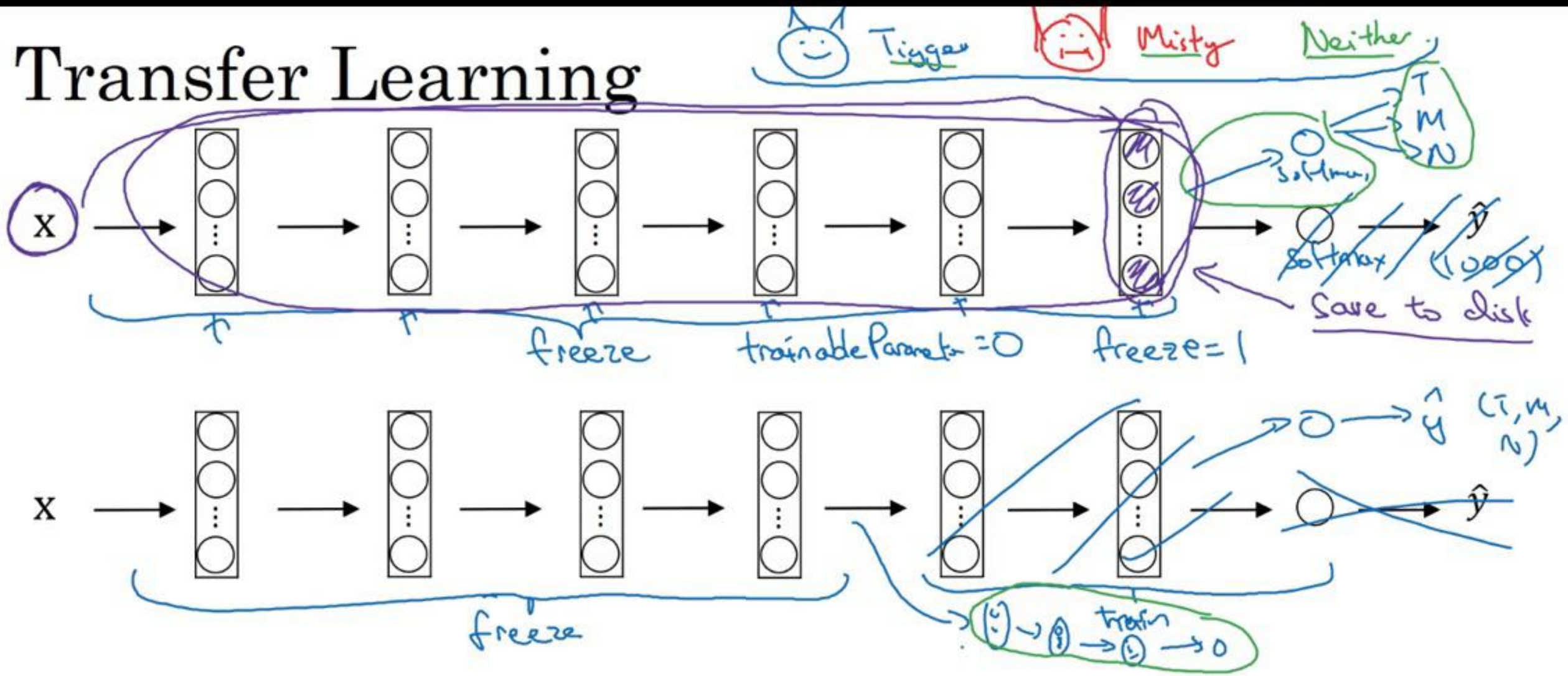
Transfer Learning



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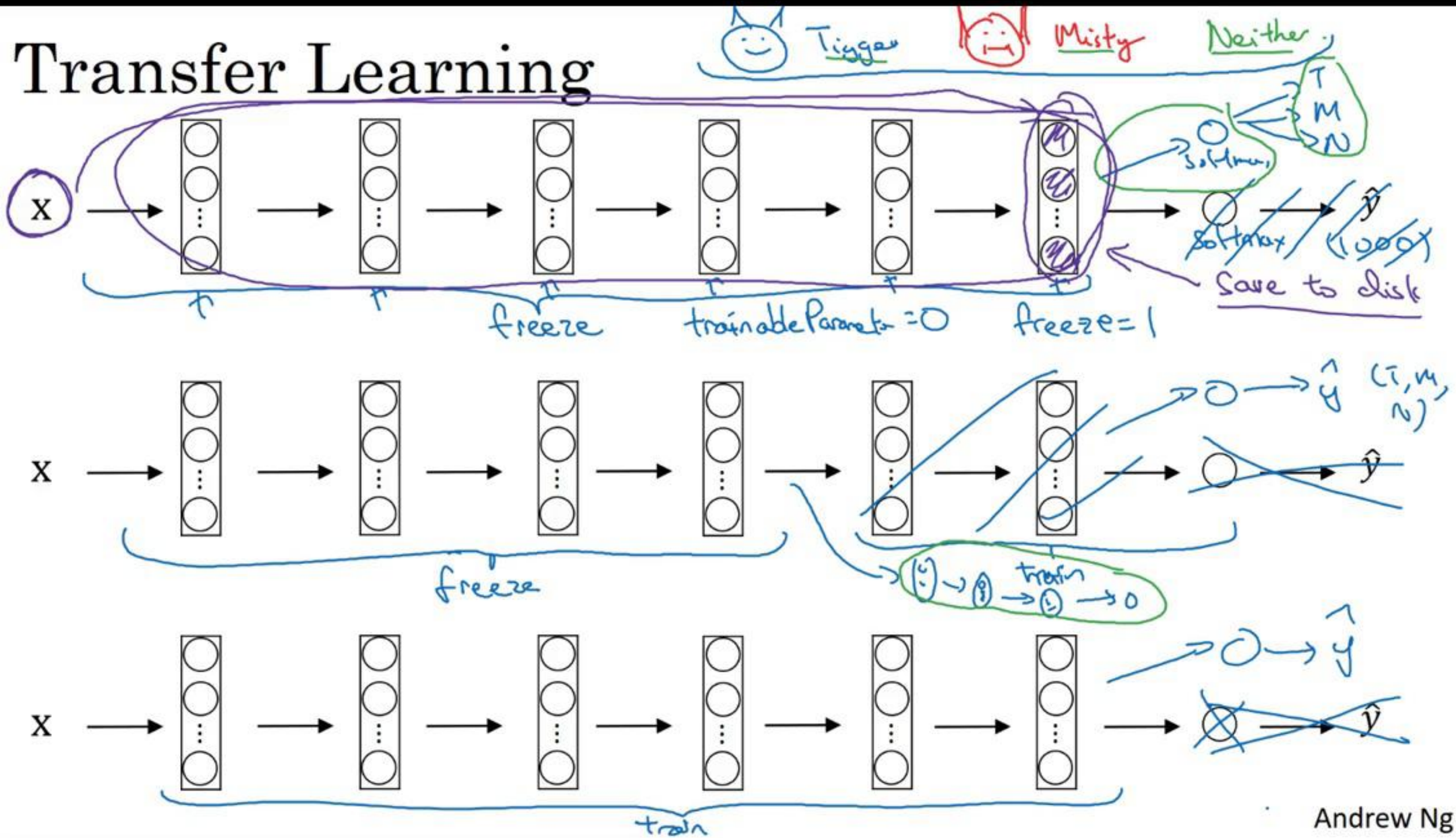
Transfer Learning



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Data augmentation

Common augmentation method

Mirroring

y

Random Cropping

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Common augmentation method

Mirroring

y

Random Cropping

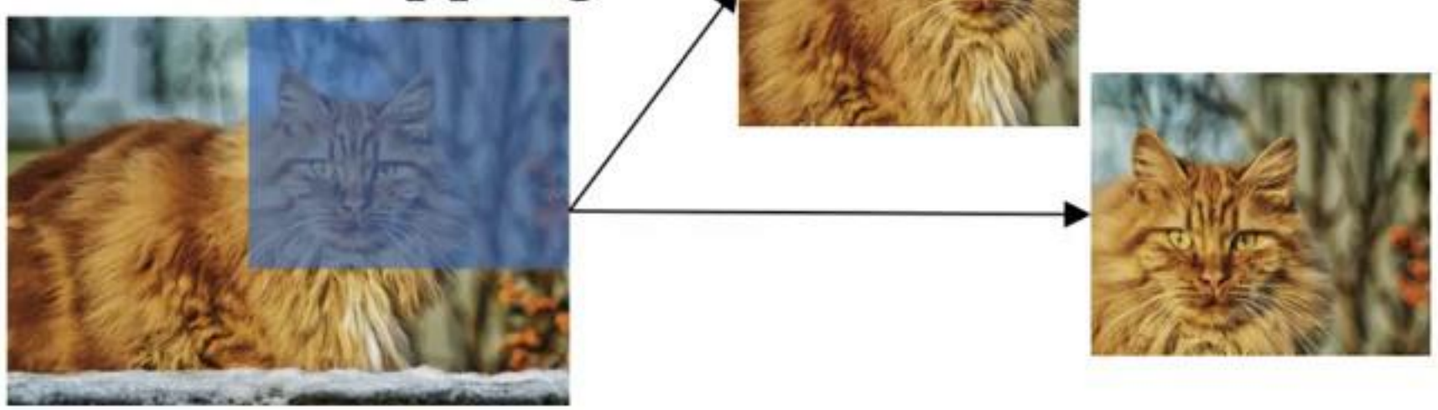
Andrew Ng

Common augmentation method

Mirroring

y

Random Cropping



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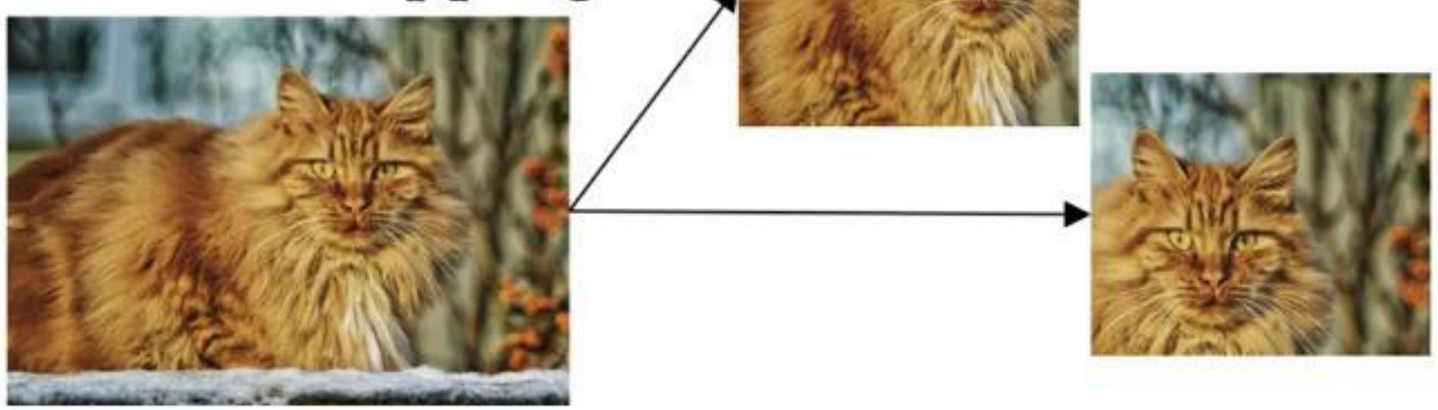
Common augmentation method

Mirroring

y

Random Cropping



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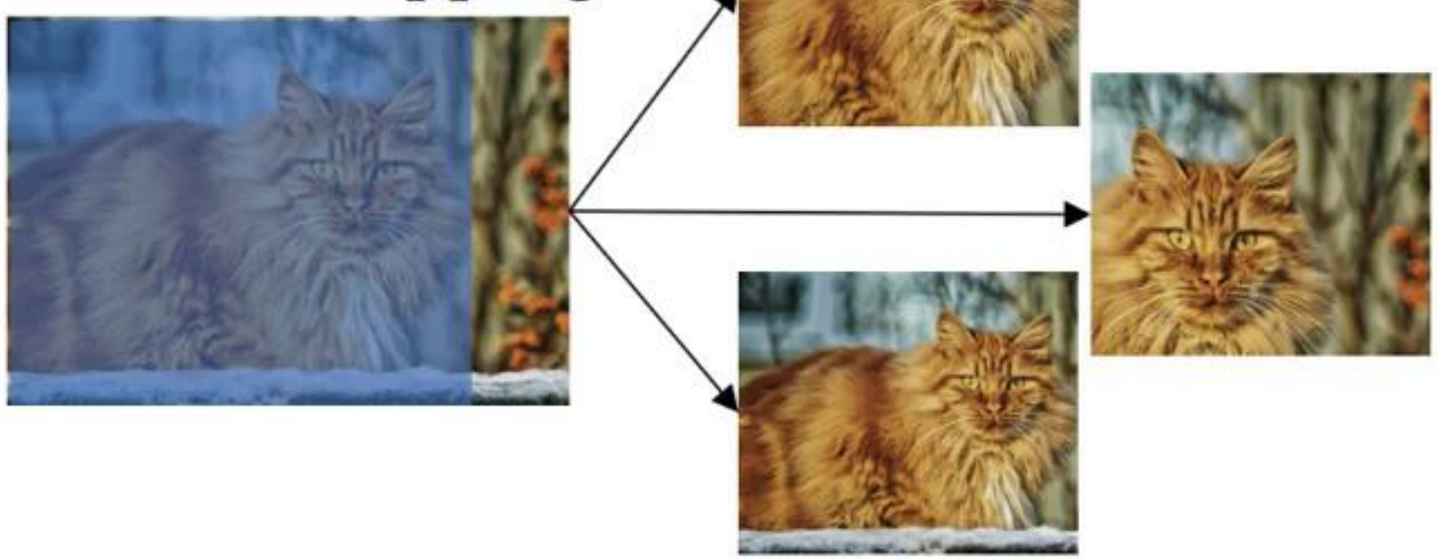
Common augmentation method

Mirroring

y

Random Cropping



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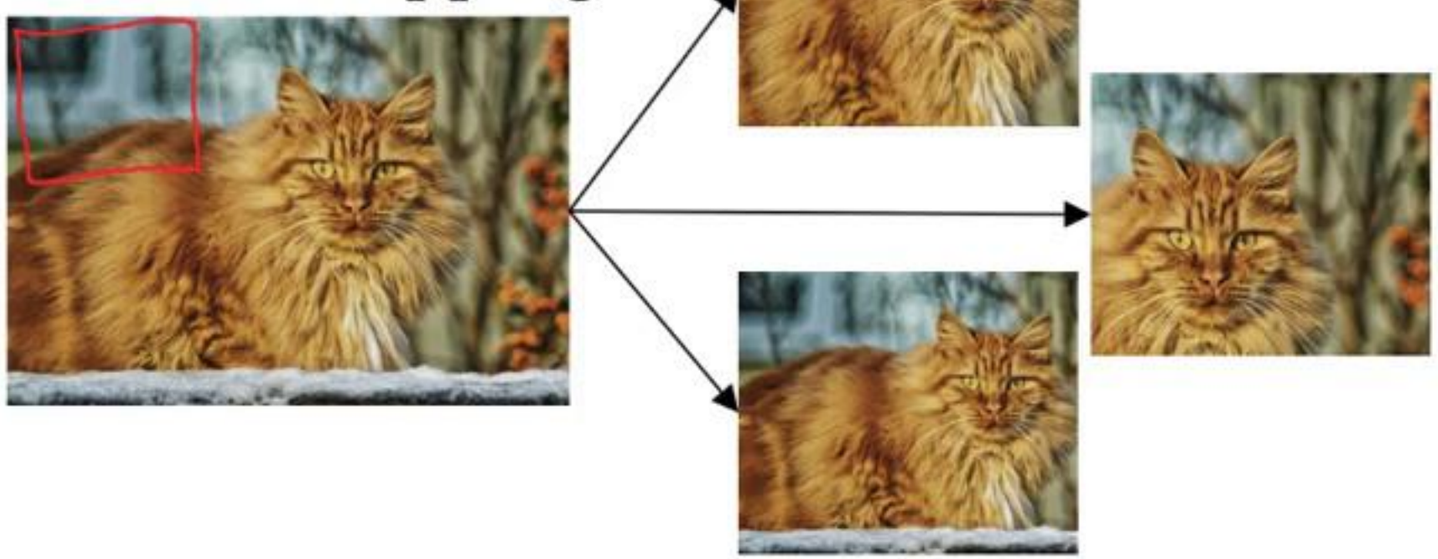
Common augmentation method

Mirroring

y

Random Cropping



- Rotation
- Shearing
- Local warping
- ...

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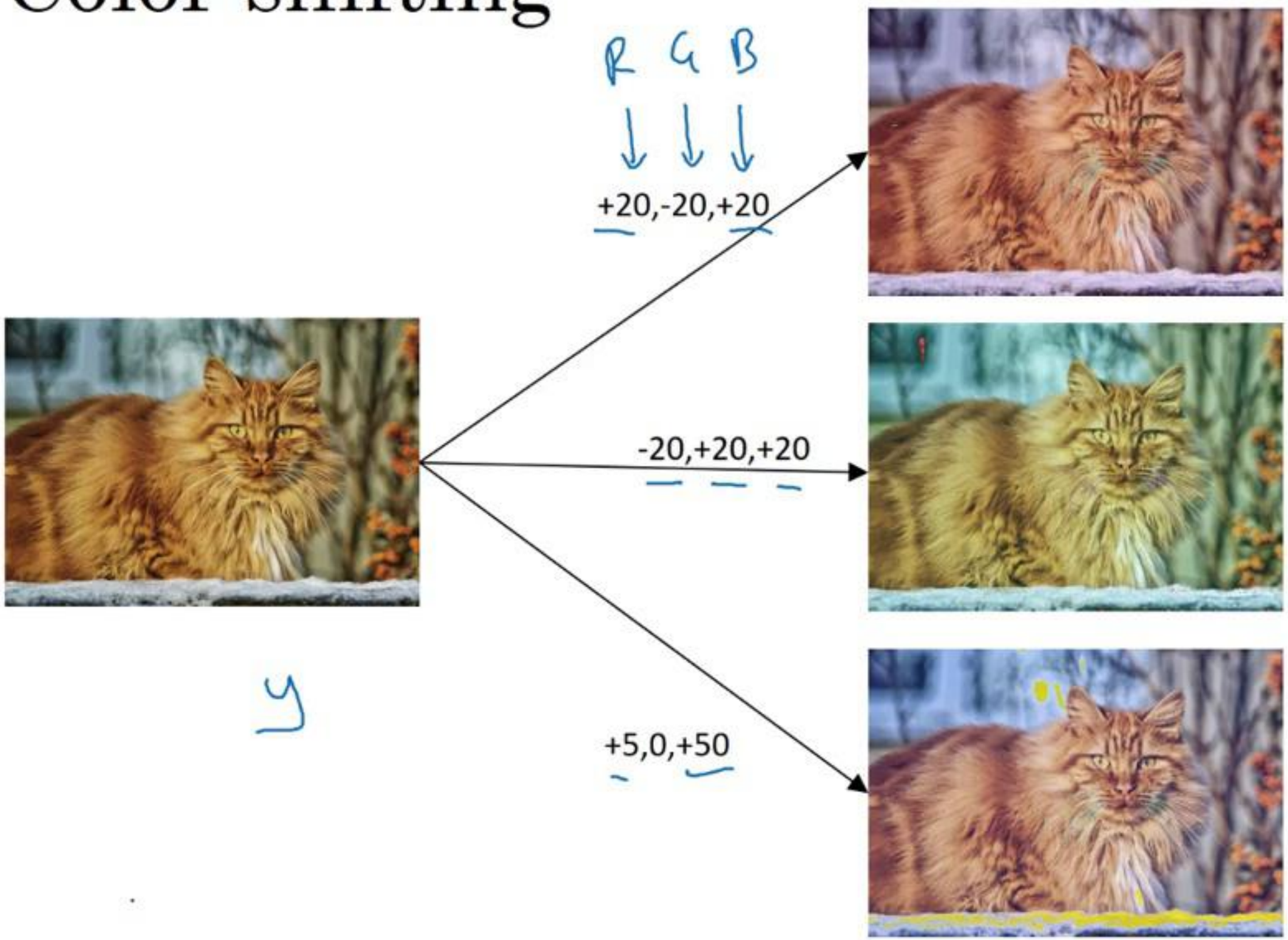
Color shifting



Advanced:

PCA

ml-class.org

[AlexNet paper

["PCA color augmentation."

R B G

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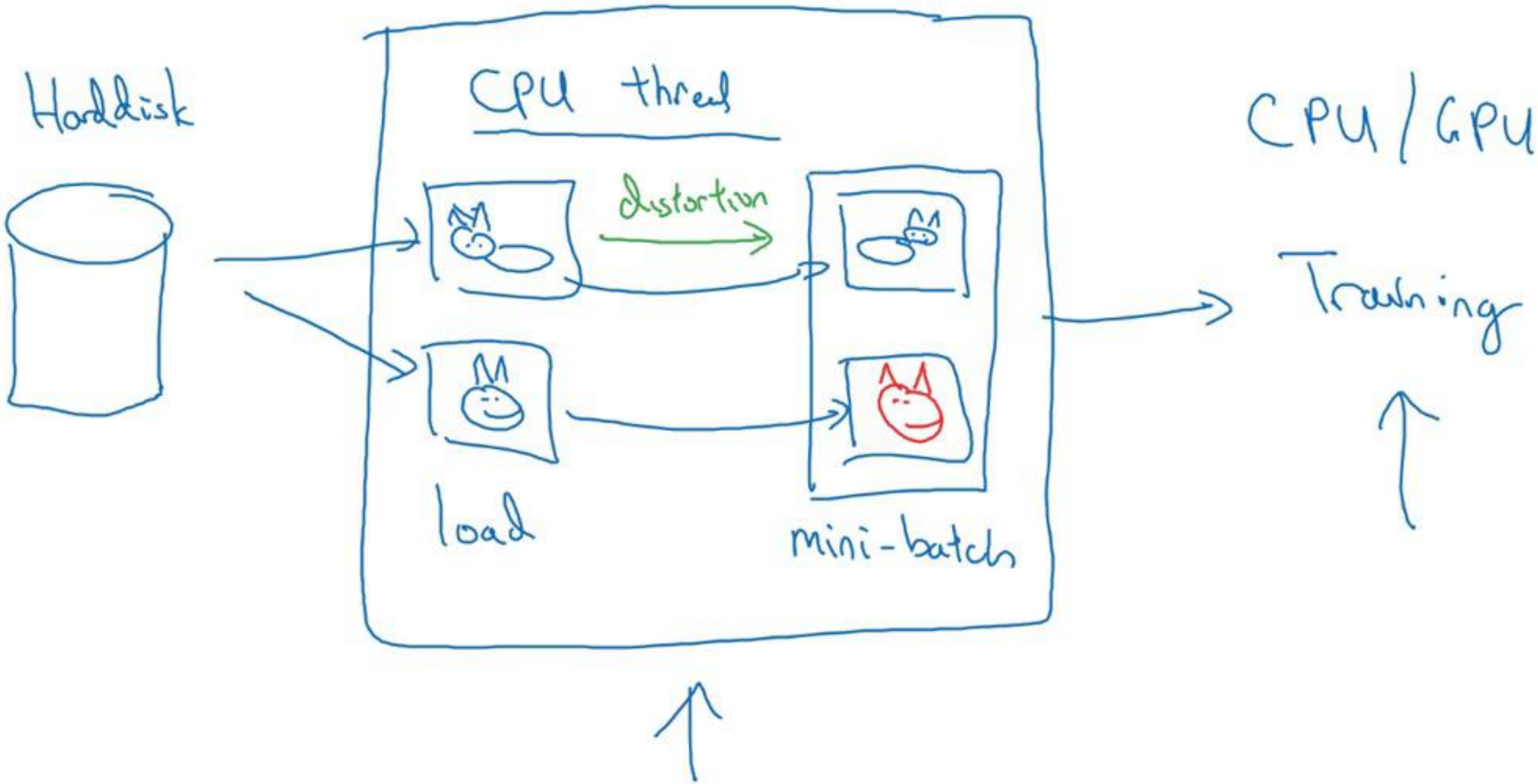
Implementing distortions during training



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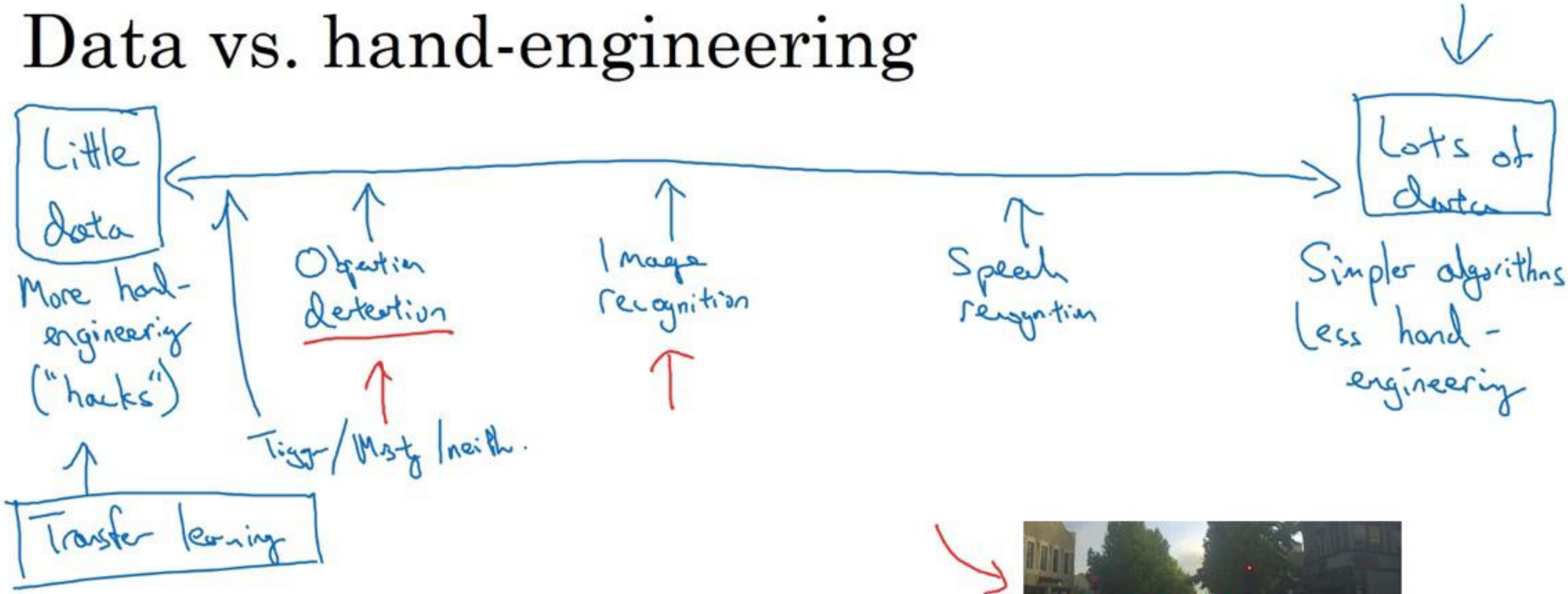
Practical advice for using ConvNets

deeplearning.ai

The state of computer vision

Data vs. hand-engineering



Two sources of knowledge

- Labeled data (x, y)
- Hand engineered features/network architecture/other components

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Tips for doing well on benchmarks/winning competitions

Ensembling

3-15 networks

$\rightarrow \hat{y}$

- Train several networks independently and average their outputs

Multi-crop at test time

- Run classifier on multiple versions of test images and average results

10-crop

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1 + 4 + 1 + 4

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Use open source code

- Use architectures of networks published in the literature
- Use open source implementations if possible
- Use pretrained models and fine-tune on your dataset